# Process documentation: before and with the system

For each of the 25 modules, how the clinic did the work before (AS-IS, from the clinic interview only) and how the Multi-Branch Dental Clinic Operation System does it now (TO-BE, from the code and its tests).

Version of 2026-10-19 · branch feat/auth-messaging-paymongo · sources: docs/defense/INTERVIEW_EVIDENCE.md, docs/defense/MODULES_FINAL.md, docs/defense/AUTOMATION_EVIDENCE.md

# Scope and how to read this document

- **AS-IS** steps come only from the interview: each shows its source (Q = first interview, PI = follow-up of 2026-10-11, MC = team confirmation of 2026-10-12). A step the interview did not describe is a **dashed amber box "to confirm with clinic"**; its question is in SECOND_INTERVIEW_QUESTIONS.md. Nothing about the old process is invented.
- **TO-BE** steps are what the system does today. Planned work is dashed and tagged "Planned"; browser-only prototypes are tagged "Mock only".
- No time savings are claimed: the interview gave no task durations, so every estimate says "not measured".
- Statuses: Implemented 14, Partial 9, Mock only 0, Planned 2.

Step from the interview (citation shown) · To confirm with the clinic · The system does it · A person does it in the system · Mock only · Planned (not built)

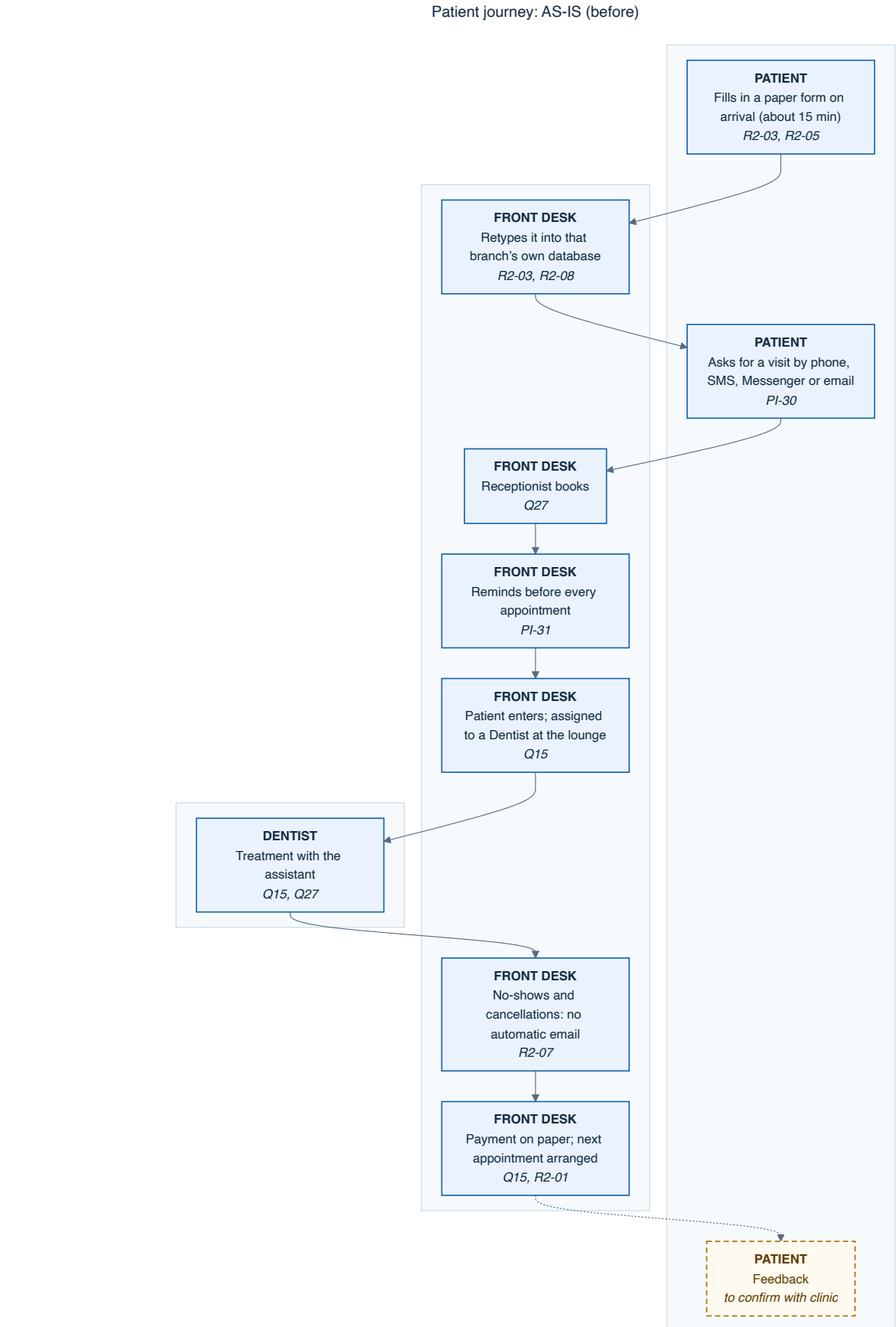
## Module summary

| # | Module | Status | Automation type |
|---|---|---|---|
| M1 | User Management | Partial | Event-driven (code and invitation emails), rule-based (role checks) |
| M2 | Branch Management | Partial | Rule-based |
| M3 | Work Schedules | Implemented | Rule-based |
| M4 | Patient Records | Partial | Rule-based (audit, name snapshot) |
| M5 | Treatment Management | Implemented | Rule-based (lock), event-driven (to billing) |
| M6 | Appointment Management | Implemented | Rule-based (open times, no overlaps), event-driven (emails), scheduled (reminders, waitlist) |
| M7 | AI Chatbot (Clinic Assistant) | Implemented | Rule-based (knowledge matching) |
| M8 | Patient Visits | Partial | Rule-based, scheduled (no-show flag; a person decides) |
| M9 | Patient Queue | Partial | Rule-based (numbering) |
| M10 | Dental Charts | Implemented | None (the system never diagnoses or suggests) |
| M11 | Billing Management | Implemented | Event-driven (draft, emails), rule-based (immutable charges), scheduled (overdue reminders) |
| M12 | Treatment Plans | Planned | Planned |
| M13 | Services and Pricing | Partial | Rule-based (effective price) |
| M14 | Document Management | Partial | Rule-based |
| M15 | Leave Management | Implemented | Rule-based |
| M16 | Patient Inquiries | Partial | Rule-based |
| M17 | Patient Messaging | Implemented | Event-driven (real-time, alert email) |
| M18 | Notifications | Implemented | Event-driven, scheduled (reminders, retries) |
| M19 | Prescriptions | Implemented | Rule-based (numbering, lock, notice) |
| M20 | Patient Follow-Up | Implemented | Rule-based tasks; optional scheduled reminder (off by default) |
| M21 | Reports and Analytics | Partial | Rule-based |
| M22 | Audit Logs | Implemented | Event-driven |
| M23 | Payment Management | Implemented | Integration (PayMongo), event-driven (receipt) |
| M24 | Lab Cases | Planned | Planned |
| M25 | Patient Handovers | Implemented | Rule-based |

# End-to-end patient journey: before (AS-IS)

Step from the interview (citation shown) · To confirm with the clinic · The system does it · A person does it in the system · Mock only · Planned (not built)

## AS-IS (from the interview)

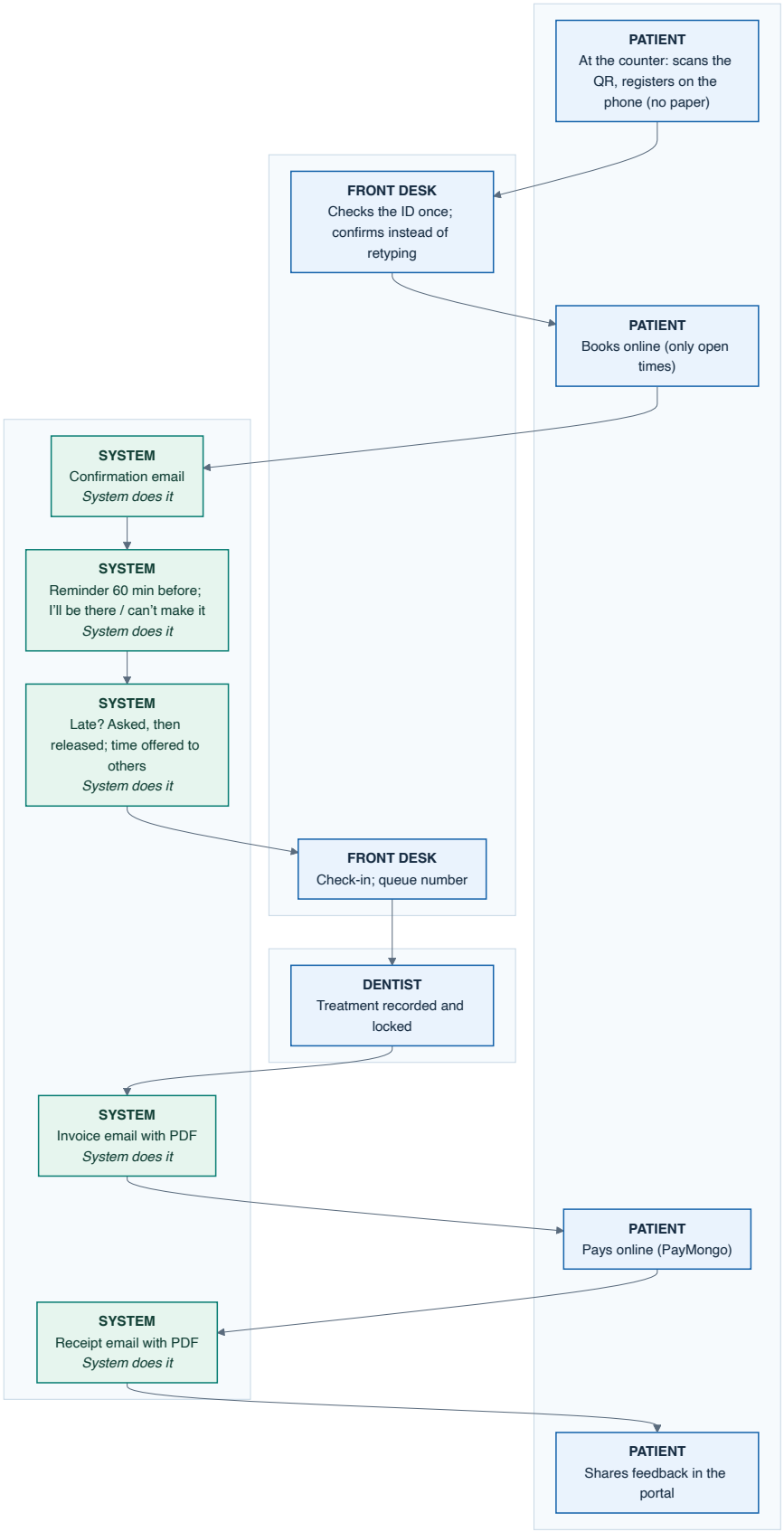


# End-to-end patient journey: with the system (TO-BE)

Step from the interview (citation shown) · To confirm with the clinic · The system does it · A person does it in the system · Mock only · Planned (not built)

## TO-BE (what the system does today)

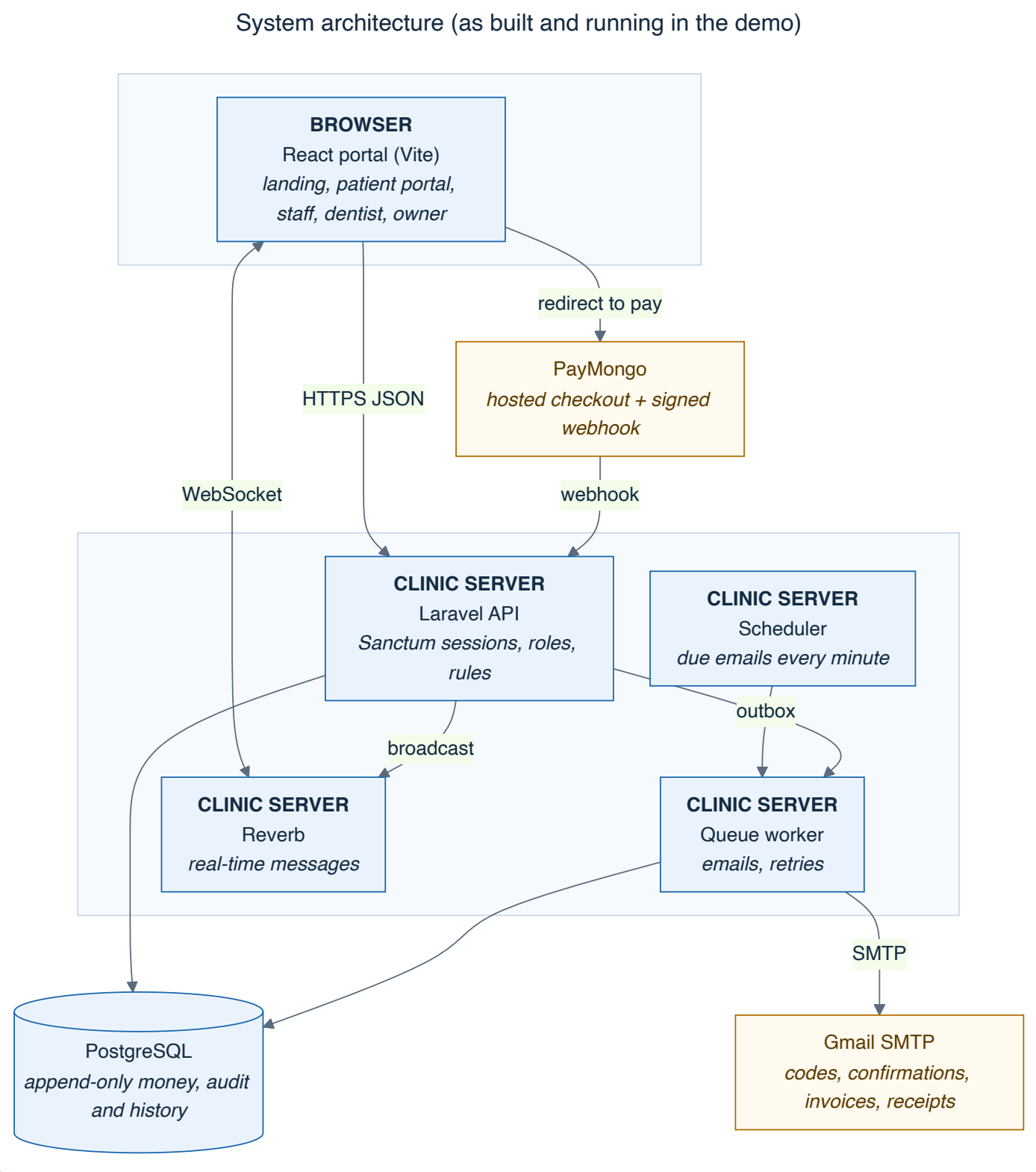
Patient journey: TO-BE (with the system)



# System architecture (as built)

Running today in the demo: one clinic server, PostgreSQL, Gmail SMTP and PayMongo test mode.

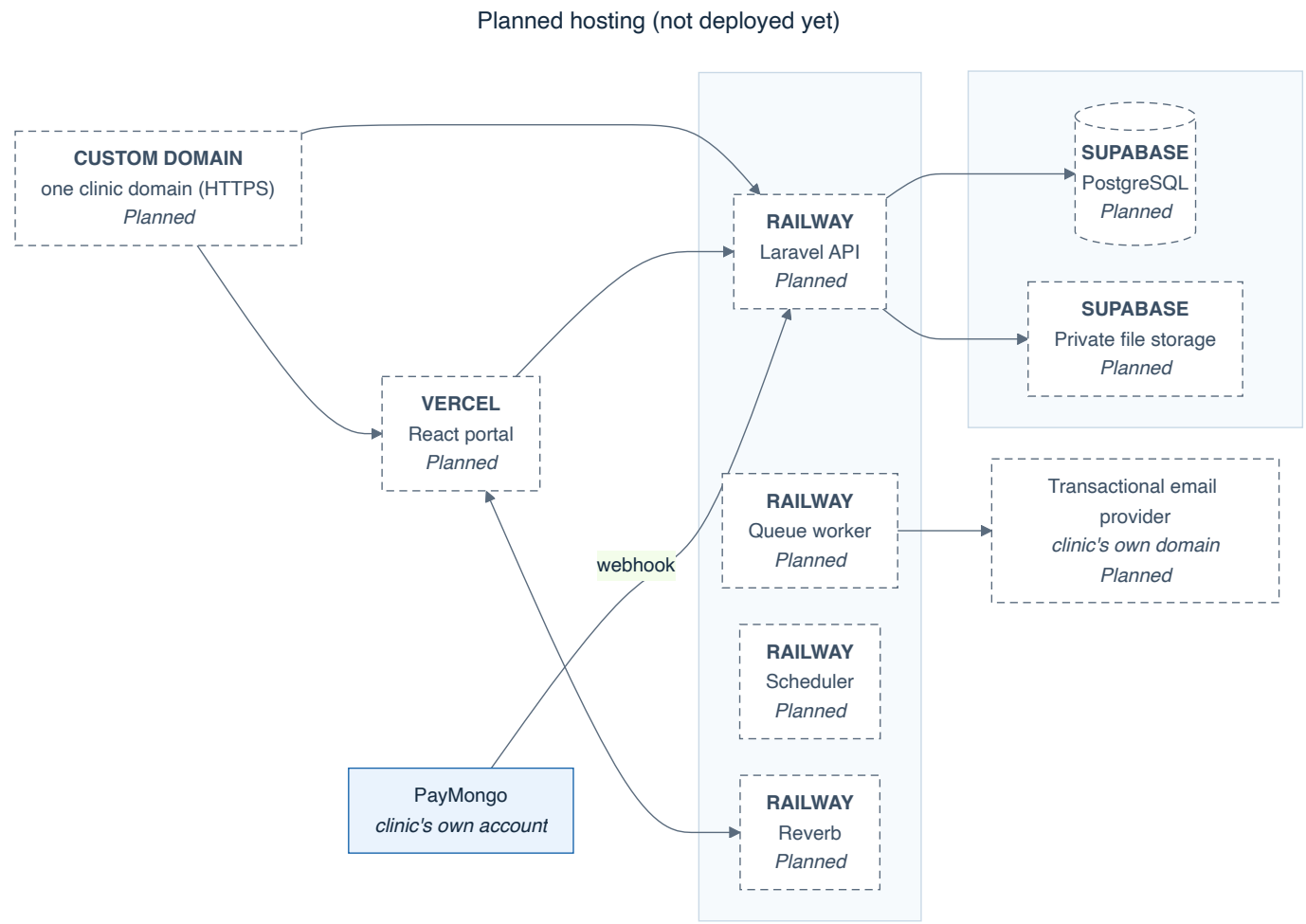
**As built and running in the demo**



# Planned hosting

Not deployed. Dashed boxes are planned; PayMongo would use the clinic's own account.

**Planned hosting (not deployed)**



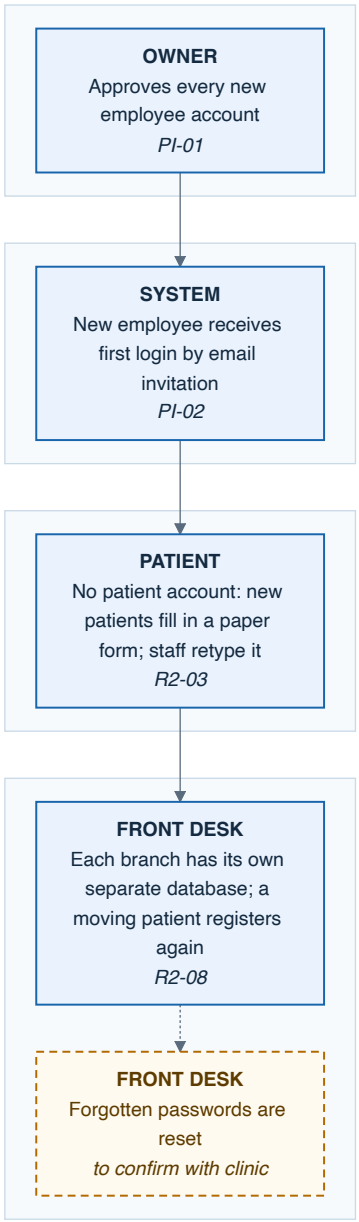
# M1 User Management

Partial

AS-IS sources: PI-01, PI-02, R2-03, R2-08 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

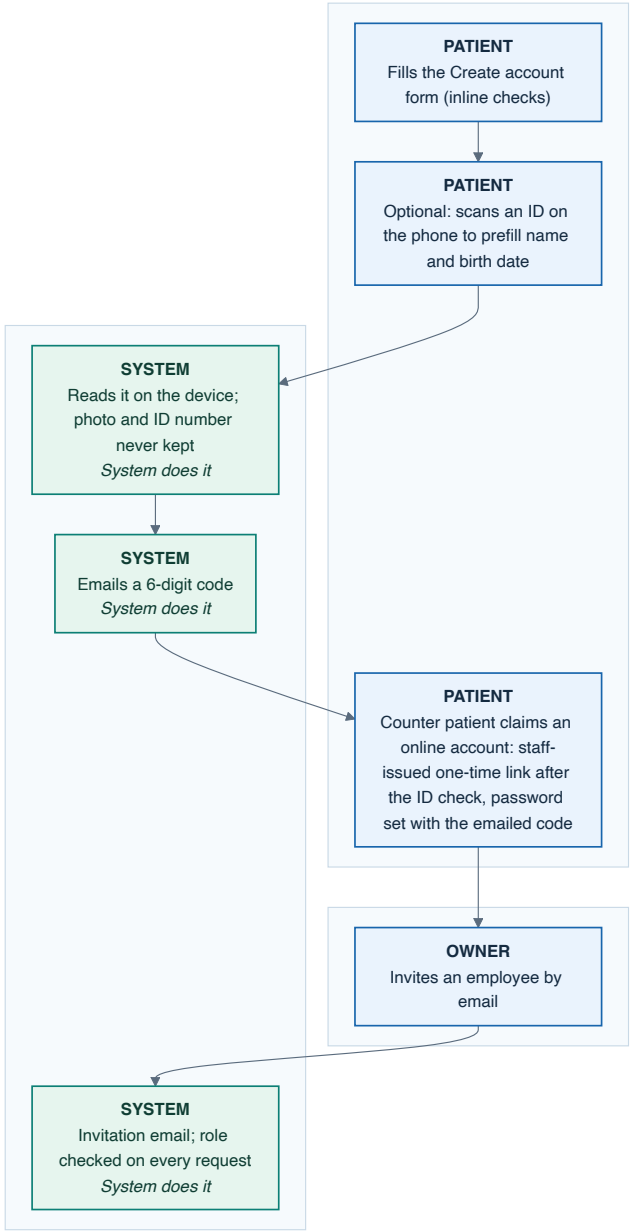
## AS-IS (before)

M1 User Management: AS-IS (before)




## TO-BE (with the system)

M1 User Management: TO-BE (with the system)




| | Before | Now |
|---|---|---|
| **Steps** | 5 known (1 to confirm) | 7 in the system |
| **Who does it** | Patient, Front desk, Owner, System | Patient, Owner, System |
| **Steps the system does by itself** | — | 3 |
| **Estimated time** | About 15 minutes for a new patient's paper sign-in (R2-05) | To be measured in pilot |
| **Errors removed** | — | Accounts are created only after the email is proven; no shared logins; one record for all branches (R2-08) |
| **Automation type** | Manual | Event-driven (code and invitation emails), rule-based (role checks) |

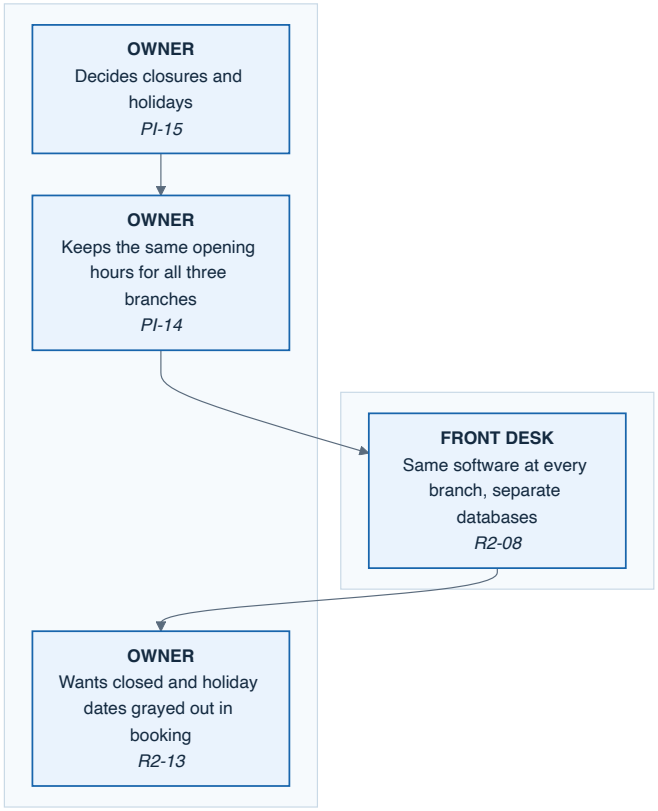
# M2 Branch Management

Partial

AS-IS sources: PI-14, PI-15, R2-08, R2-13

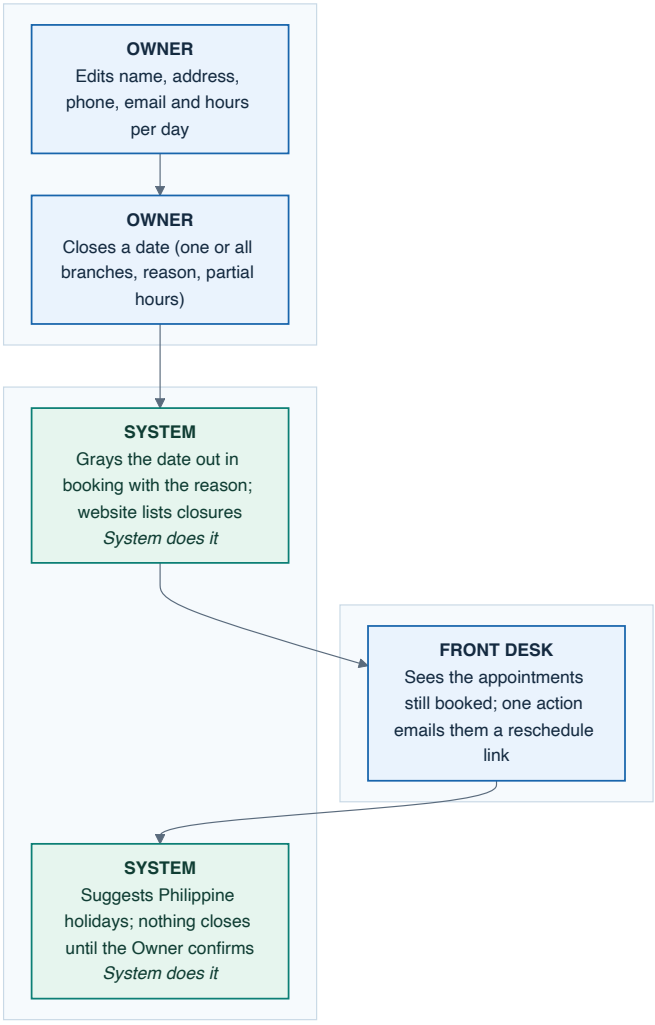
## AS-IS (before)

M2 Branch Management: AS-IS (before)




## TO-BE (with the system)

M2 Branch Management: TO-BE (with the system)



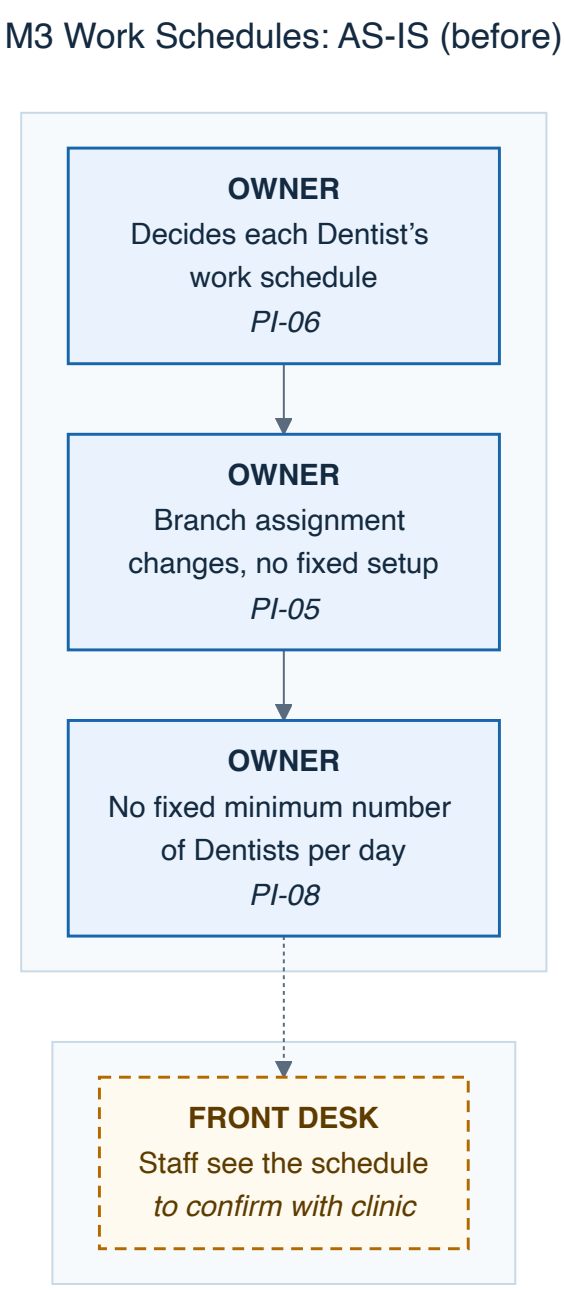

| | Before | Now |
|---|---|---|
| **Steps** | 4 known | 5 in the system |
| **Who does it** | Front desk, Owner | Front desk, Owner, System |
| **Steps the system does by itself** | — | 2 |
| **Estimated time** | Not measured (no figure in the interviews) | To be measured in pilot |
| **Errors removed** | — | Old or different hours in different places; bookings on closed days |
| **Automation type** | Manual | Rule-based |

# M3 Work Schedules

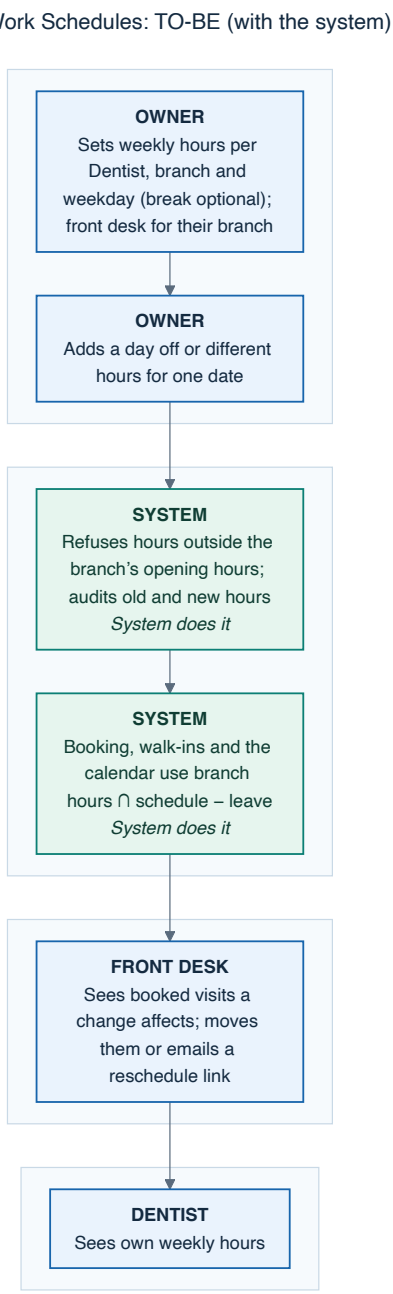
Implemented

AS-IS sources: PI-05, PI-06, PI-08 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

**AS-IS (before)**




**TO-BE (with the system)**




| | Before | Now |
|---|---|---|
| **Steps** | 4 known (1 to confirm) | 6 in the system |
| **Who does it** | Front desk, Owner | Front desk, Dentist, Owner, System |
| **Steps the system does by itself** | — | 2 |
| **Estimated time** | Not measured (no figure in the interviews) | To be measured in pilot |
| **Errors removed** | — | Booking outside a Dentist's hours; a change silently leaving patients without a Dentist |
| **Automation type** | Manual | Rule-based |

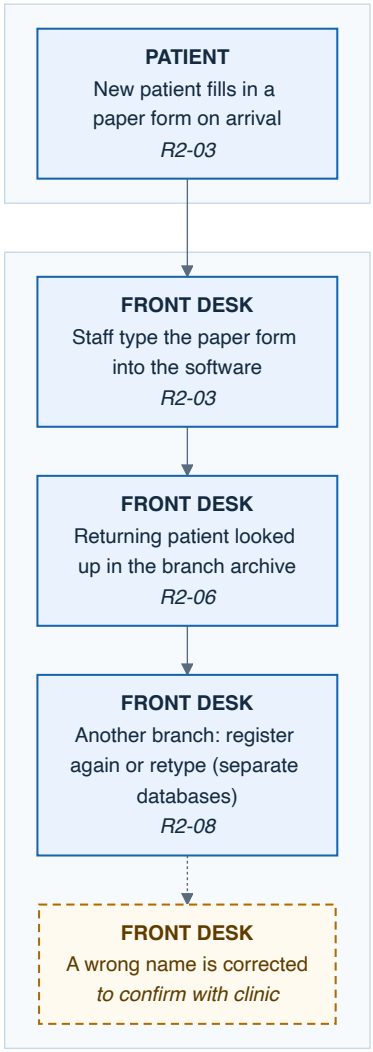
# M4 Patient Records

Partial

AS-IS sources: R2-03, R2-06, R2-08 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

## AS-IS (before)

M4 Patient Records: AS-IS (before)

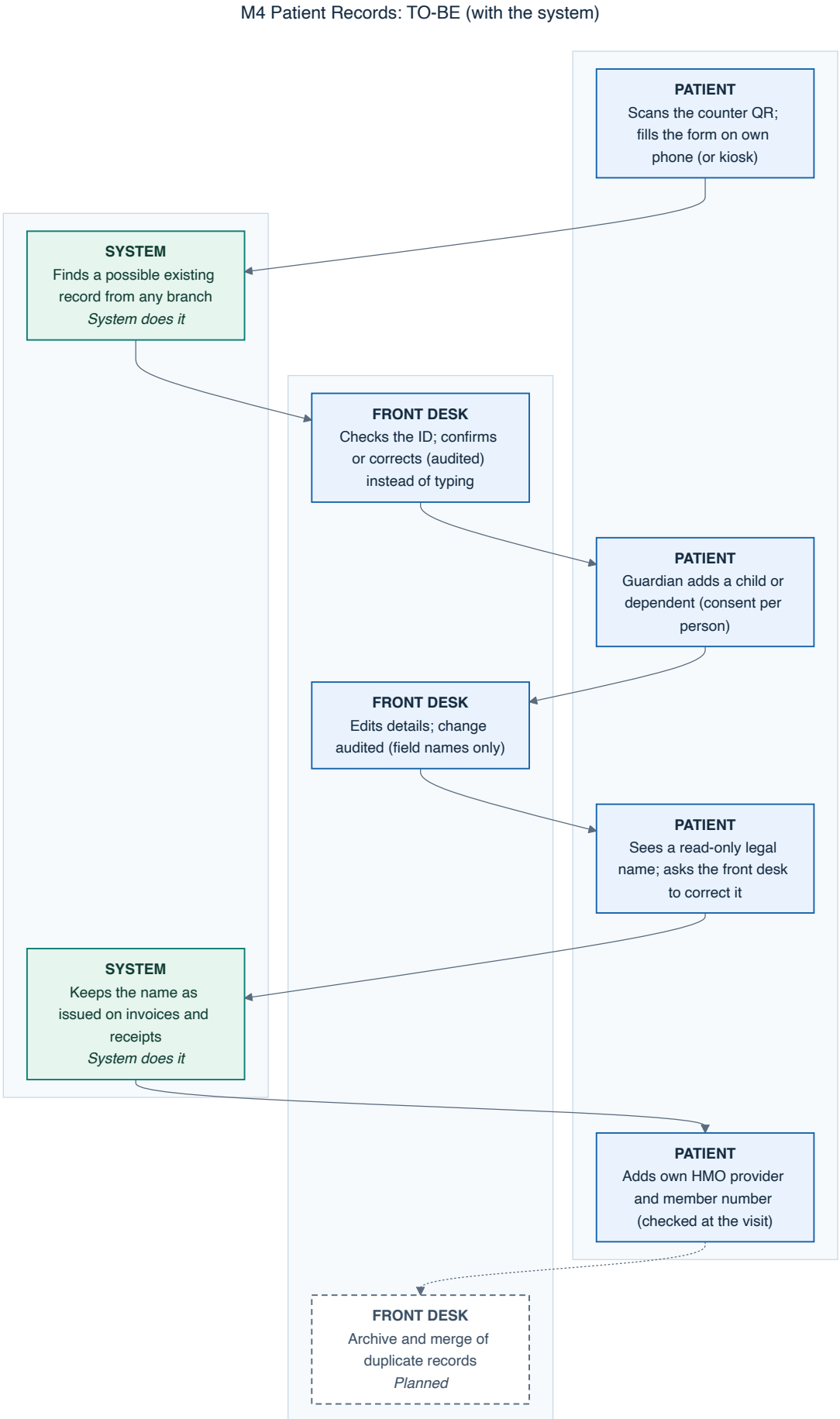



| | Before | Now |
|---|---|---|
| **Steps** | 5 known (1 to confirm) | 8 in the system + 1 planned or mock |
| **Who does it** | Patient, Front desk | Patient, Front desk, System |
| **Steps the system does by itself** | — | 2 |
| **Estimated time** | About 15 minutes of paper sign-in, then retyping (R2-03, R2-05) | Phone self-registration plus a staff check: to be measured in pilot |
| **Errors removed** | — | Retyping paper forms (R2-03) and re-registering at another branch (R2-08); untracked edits |
| **Automation type** | Manual | Rule-based (audit, name snapshot) |

# M4 Patient Records: TO-BE

Partial

## TO-BE (with the system)

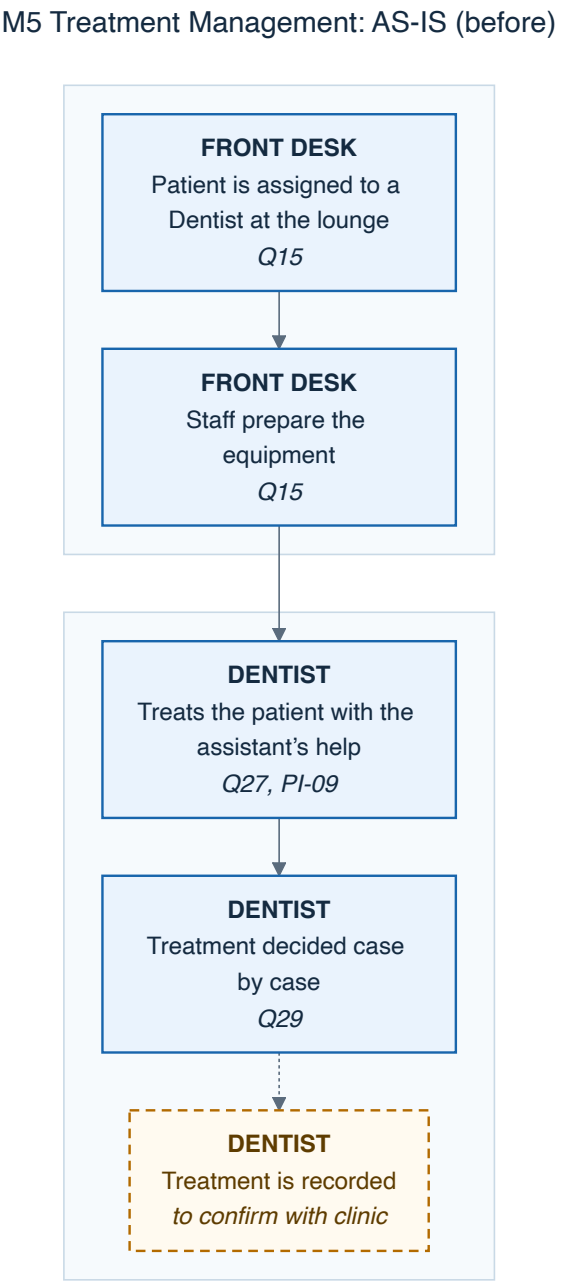


# M5 Treatment Management

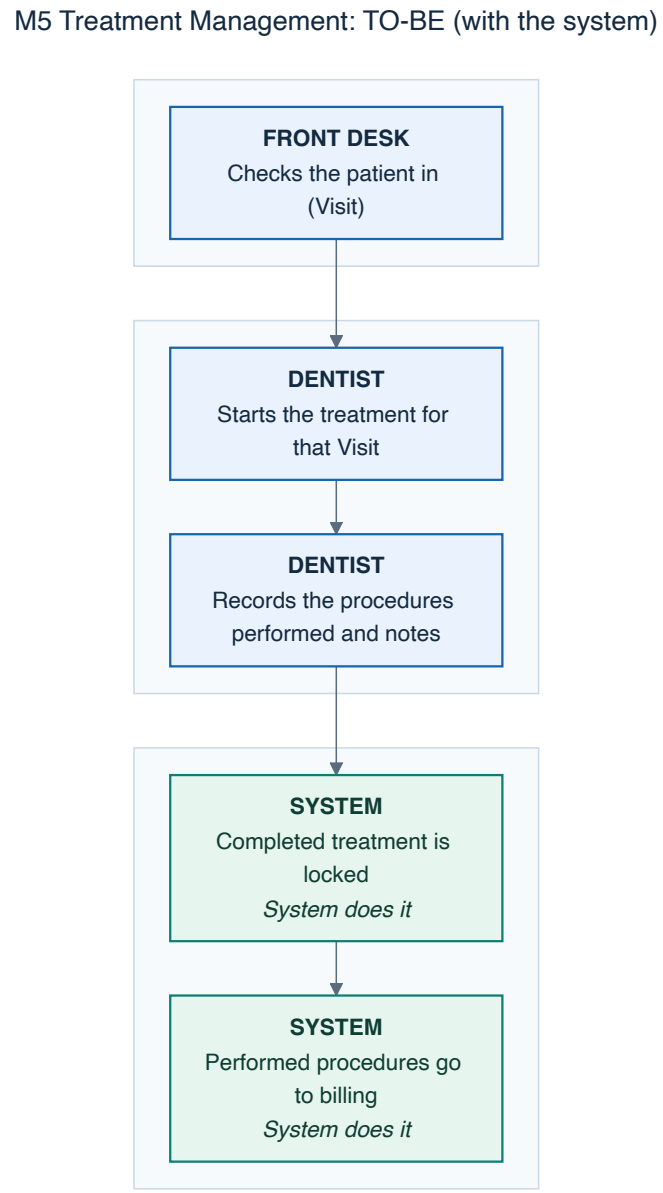
Implemented

AS-IS sources: Q15, Q27, PI-09, Q29 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

**AS-IS (before)**




**TO-BE (with the system)**



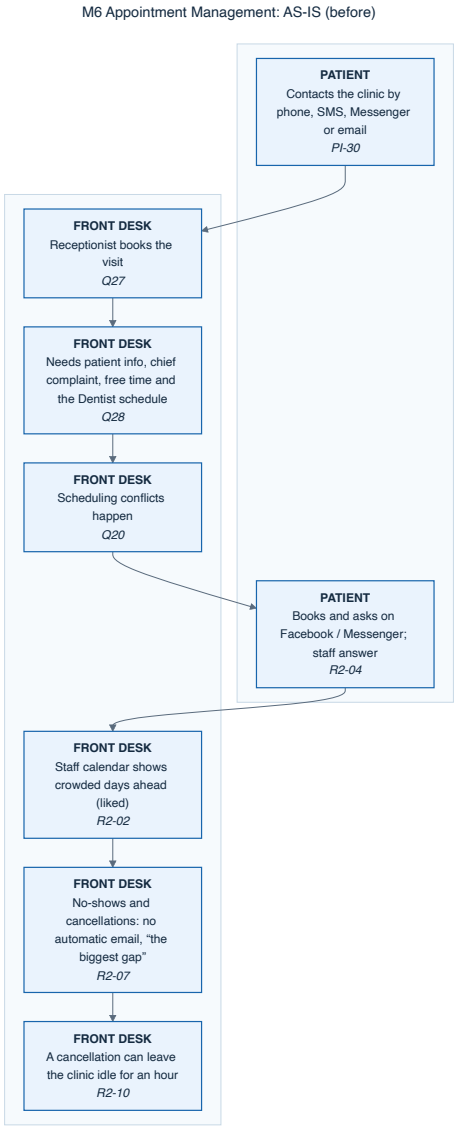

| | Before | Now |
|---|---|---|
| **Steps** | 5 known (1 to confirm) | 5 in the system |
| **Who does it** | Front desk, Dentist | Front desk, Dentist, System |
| **Steps the system does by itself** | — | 2 |
| **Estimated time** | About 20–60 minutes per procedure, depending on the treatment (R2-05) | Unchanged; to be measured in pilot |
| **Errors removed** | — | Treatment changed after completion; charges not matching the work done |
| **Automation type** | Manual | Rule-based (lock), event-driven (to billing) |

# M6 Appointment Management

Implemented

AS-IS sources: PI-30, Q20, Q27, Q28, R2-02, R2-04, R2-07, R2-10

## AS-IS (before)



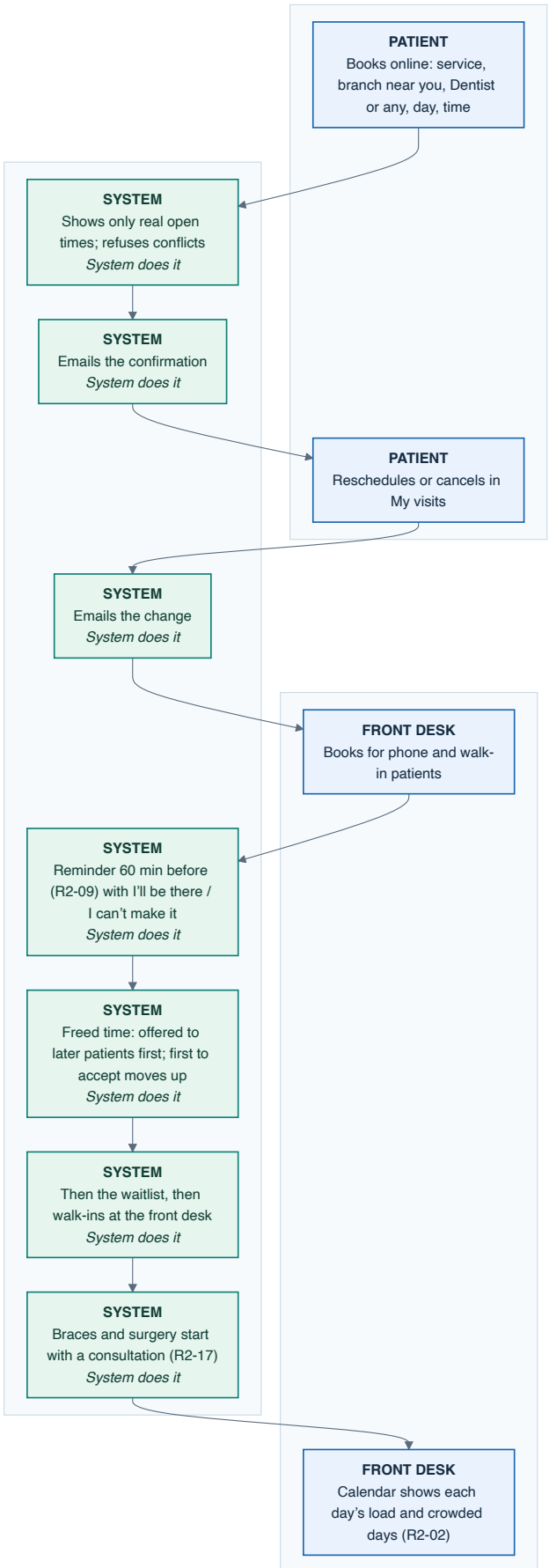

| | Before | Now |
|---|---|---|
| **Steps** | 8 known | 11 in the system |
| **Who does it** | Patient, Front desk | Patient, Front desk, System |
| **Steps the system does by itself** | — | 7 |
| **Estimated time** | Not measured; idle hour after a cancellation (R2-10) | To be measured in pilot |
| **Errors removed** | — | Scheduling conflicts (Q20) blocked by the database; no-shows (R2-07) met by reminders, one-click answers and freed-time offers |
| **Automation type** | Manual | Rule-based (open times, no overlaps), event-driven (emails), scheduled (reminders, waitlist) |

# M6 Appointment Management: TO-BE

Implemented

## TO-BE (with the system)

M6 Appointment Management: TO-BE (with the system)

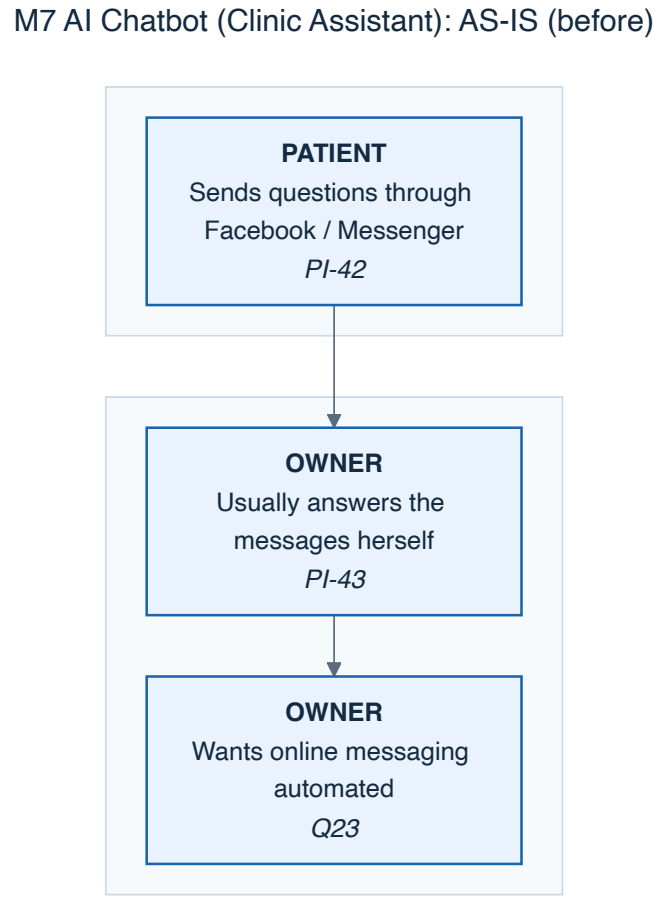


# M7 AI Chatbot (Clinic Assistant)

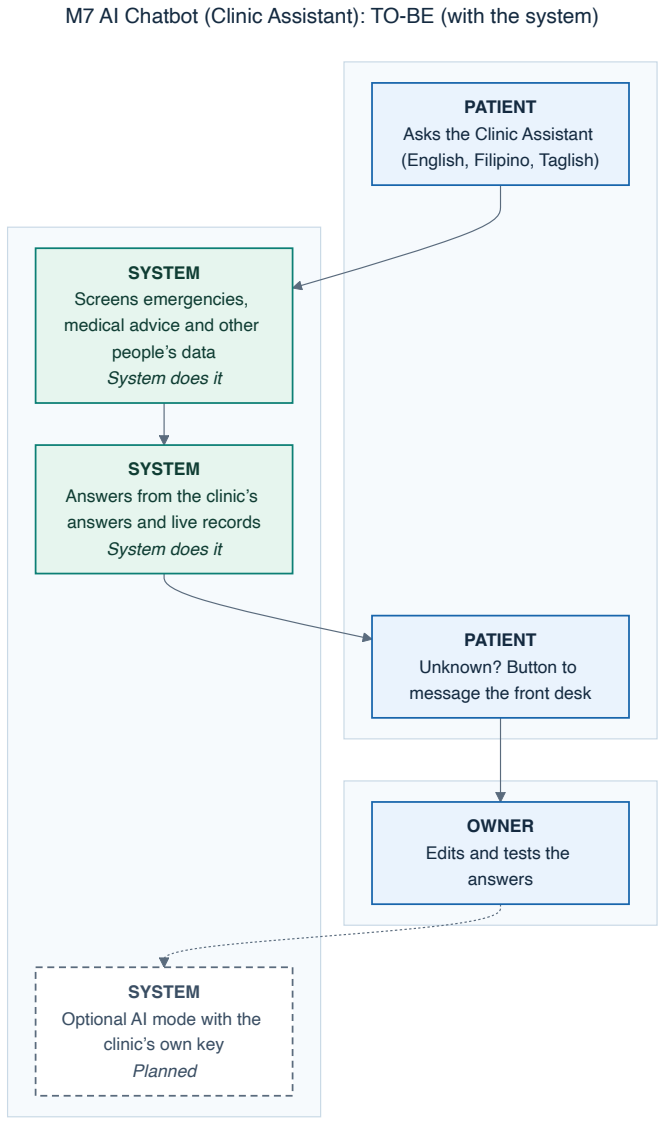
Implemented

AS-IS sources: PI-42, PI-43, Q23

## AS-IS (before)




## TO-BE (with the system)




| | Before | Now |
|---|---|---|
| **Steps** | 3 known | 5 in the system + 1 planned or mock |
| **Who does it** | Patient, Owner | Patient, Owner, System |
| **Steps the system does by itself** | — | 2 |
| **Estimated time** | Not measured (no figure in the interviews) | To be measured in pilot |
| **Errors removed** | — | Routine questions no longer wait for the Owner |
| **Automation type** | Manual | Rule-based (knowledge matching) |

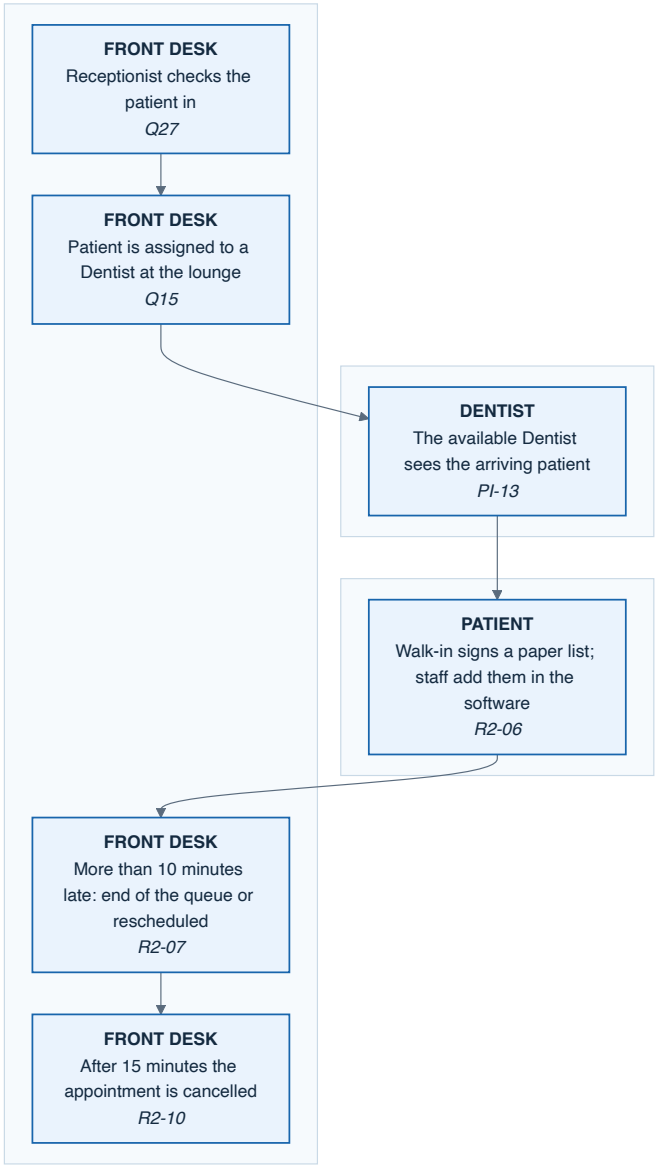
# M8 Patient Visits

Partial

AS-IS sources: PI-13, Q15, Q27, R2-06, R2-07, R2-10

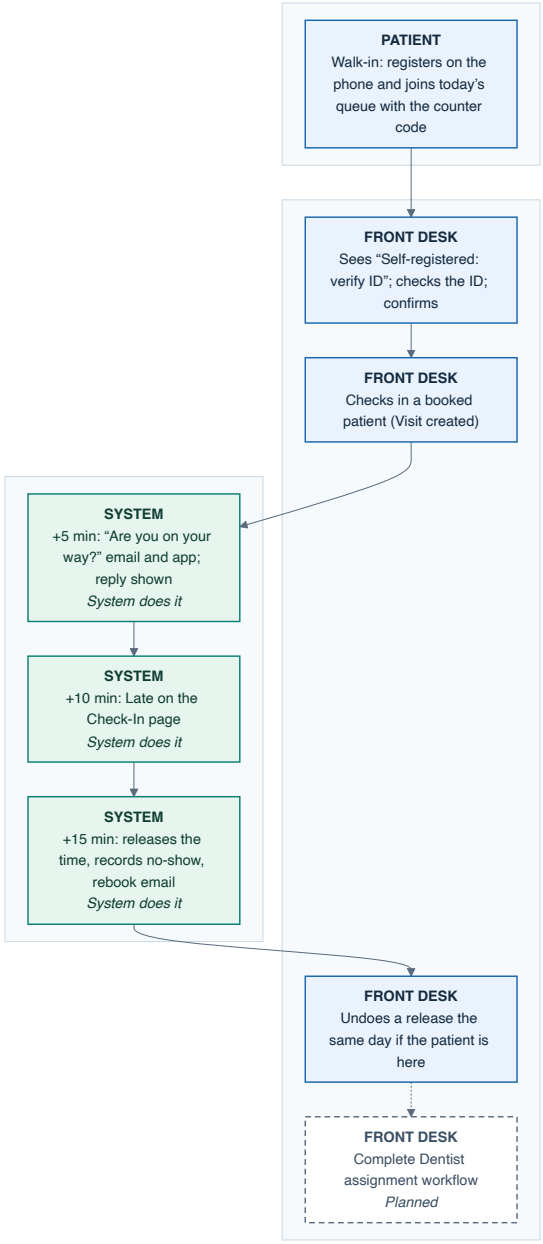
## AS-IS (before)

M8 Patient Visits: AS-IS (before)




## TO-BE (with the system)

M8 Patient Visits: TO-BE (with the system)



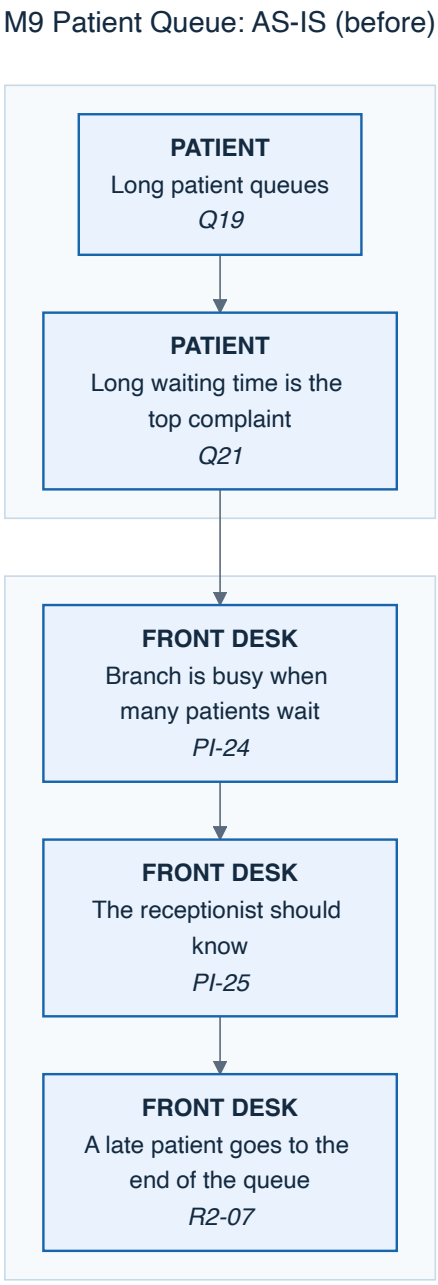

| | Before | Now |
|---|---|---|
| **Steps** | 6 known | 7 in the system + 1 planned or mock |
| **Who does it** | Patient, Front desk, Dentist | Patient, Front desk, System |
| **Steps the system does by itself** | — | 3 |
| **Estimated time** | About 15 minutes for a new patient's sign-in (R2-05) | Phone self-registration plus a staff check: to be measured in pilot |
| **Errors removed** | — | Arrivals not linked to their booking; missed visits noticed late |
| **Automation type** | Manual | Rule-based, scheduled (no-show flag; a person decides) |

# M9 Patient Queue

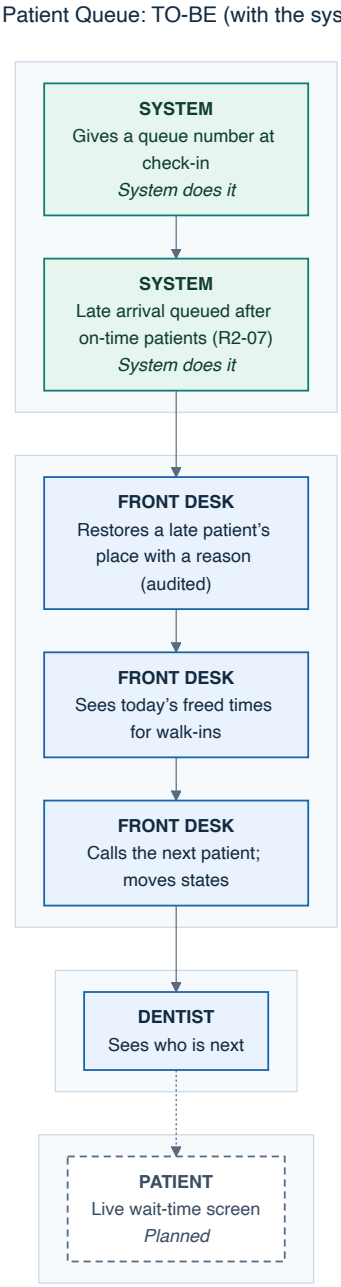
Partial

AS-IS sources: PI-24, PI-25, Q19, Q21, R2-07

### AS-IS (before)




### TO-BE (with the system)



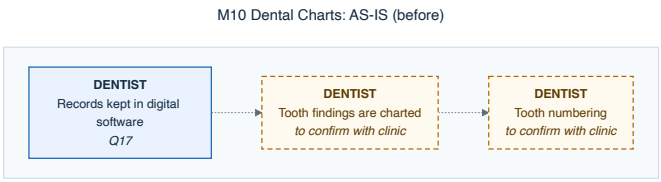

| | Before | Now |
|---|---|---|
| **Steps** | 5 known | 6 in the system + 1 planned or mock |
| **Who does it** | Patient, Front desk | Front desk, Dentist, System |
| **Steps the system does by itself** | — | 2 |
| **Estimated time** | Not measured (no figure in the interviews) | To be measured in pilot |
| **Errors removed** | — | Not measured |
| **Automation type** | Manual | Rule-based (numbering) |

# M10 Dental Charts

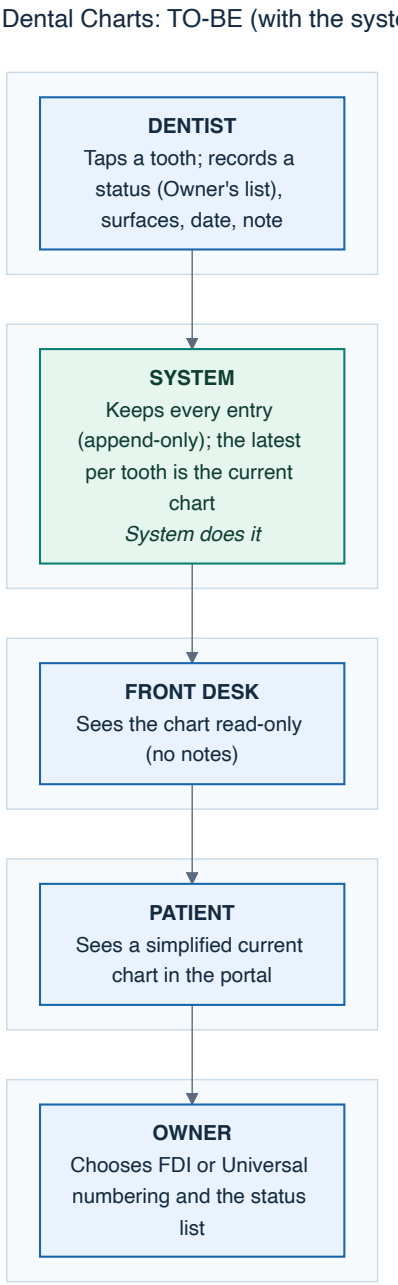
Implemented

AS-IS sources: Q17 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

## AS-IS (before)




## TO-BE (with the system)




| | Before | Now |
|---|---|---|
| **Steps** | 3 known (2 to confirm) | 5 in the system |
| **Who does it** | Dentist | Patient, Front desk, Dentist, Owner, System |
| **Steps the system does by itself** | — | 1 |
| **Estimated time** | Not measured (no figure in the interviews) | To be measured in pilot |
| **Errors removed** | — | Findings lost or overwritten; no history per tooth |
| **Automation type** | Manual | None (the system never diagnoses or suggests) |

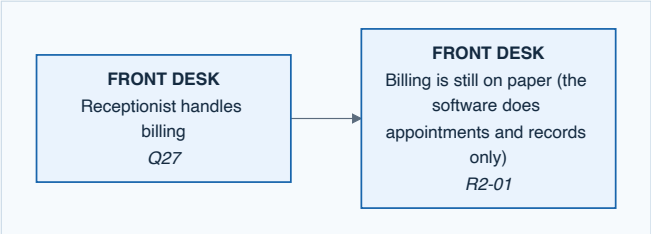
# M11 Billing Management

Implemented

AS-IS sources: Q27, R2-01

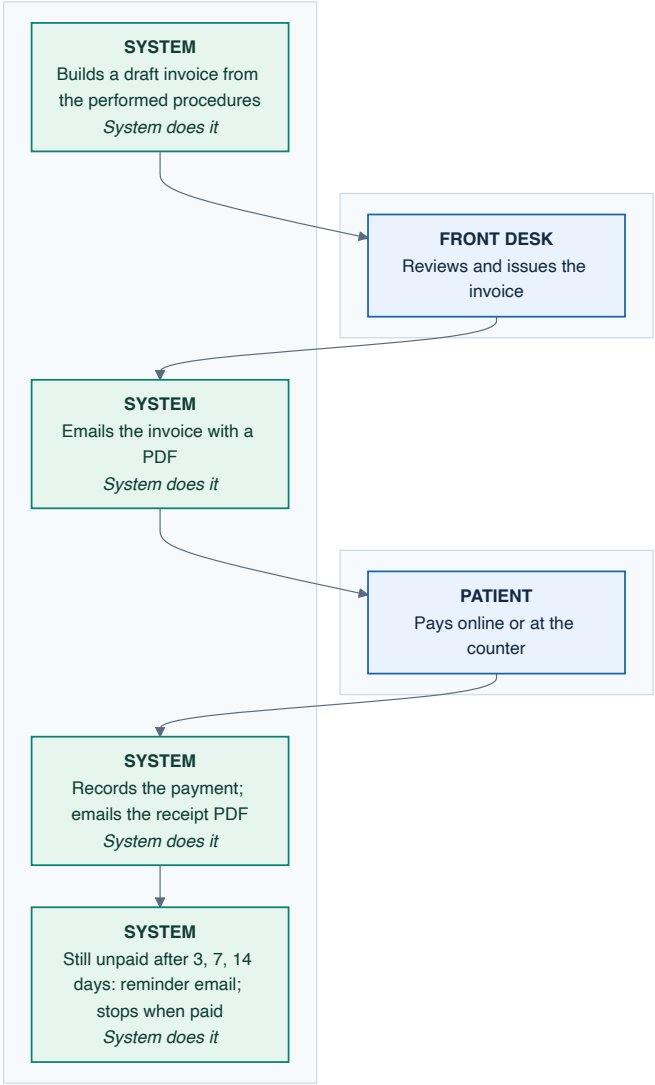
## AS-IS (before)

M11 Billing Management: AS-IS (before)




## TO-BE (with the system)

M11 Billing Management: TO-BE (with the system)



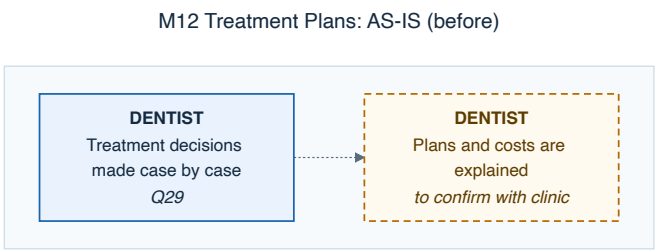

| | Before | Now |
|---|---|---|
| **Steps** | 2 known | 6 in the system |
| **Who does it** | Front desk | Patient, Front desk, System |
| **Steps the system does by itself** | — | 4 |
| **Estimated time** | Not measured (no figure in the interviews) | To be measured in pilot |
| **Errors removed** | — | Charges not matching the treatment; edited receipts (database blocks changes); unpaid balances forgotten |
| **Automation type** | Manual | Event-driven (draft, emails), rule-based (immutable charges), scheduled (overdue reminders) |

# M12 Treatment Plans

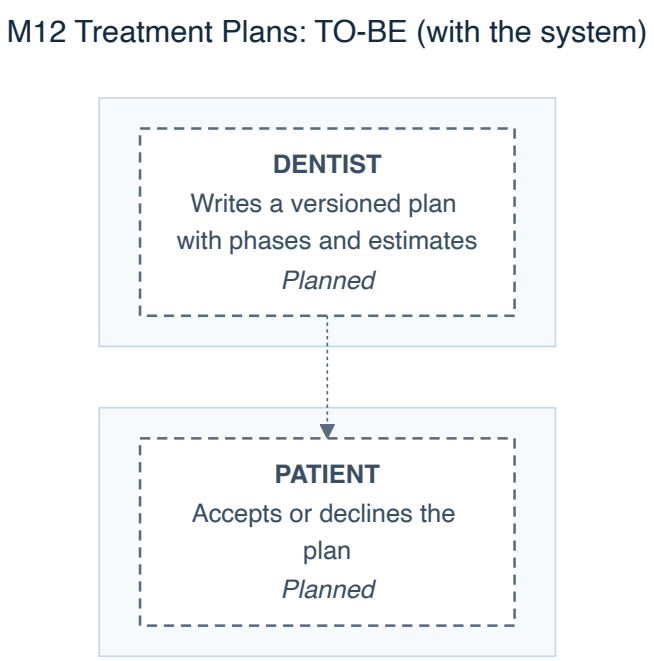
Planned

AS-IS sources: Q29 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

## AS-IS (before)




## TO-BE (with the system)



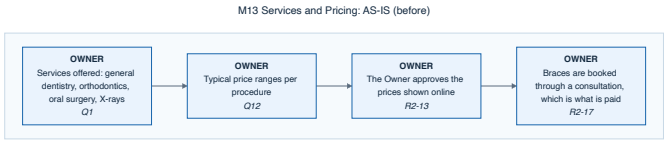

| | Before | Now |
|---|---|---|
| **Steps** | 2 known (1 to confirm) | 0 in the system + 2 planned or mock |
| **Who does it** | Dentist | Nobody yet (planned) |
| **Steps the system does by itself** | — | 0 |
| **Estimated time** | Not measured (no figure in the interviews) | To be measured in pilot |
| **Errors removed** | — | Not built yet |
| **Automation type** | Manual | Planned |

# M13 Services and Pricing

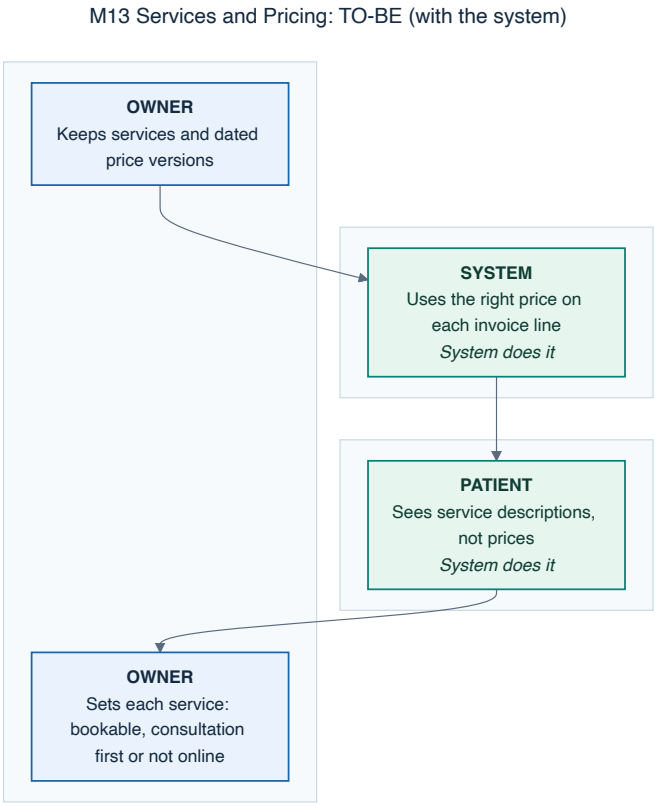
Partial

AS-IS sources: Q1, Q12, R2-13, R2-17

## AS-IS (before)




## TO-BE (with the system)




| | Before | Now |
|---|---|---|
| **Steps** | 4 known | 4 in the system |
| **Who does it** | Owner | Patient, Owner, System |
| **Steps the system does by itself** | — | 2 |
| **Estimated time** | Not measured (no figure in the interviews) | To be measured in pilot |
| **Errors removed** | — | Wrong or outdated prices on invoices |
| **Automation type** | Manual | Rule-based (effective price) |

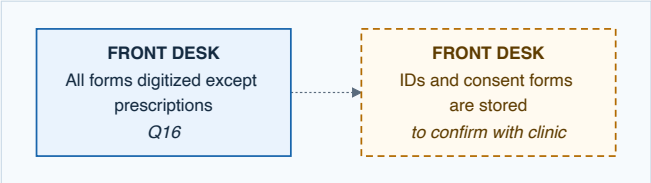
# M14 Document Management

Partial

AS-IS sources: Q16 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

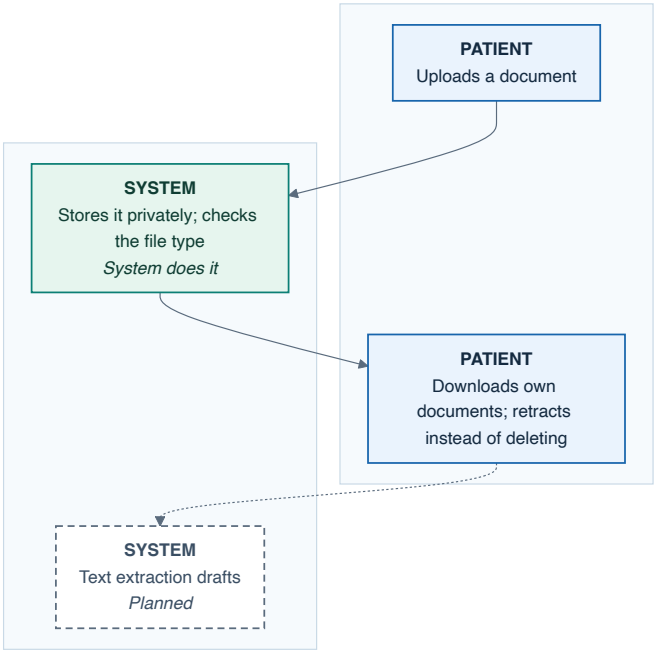
## AS-IS (before)

M14 Document Management: AS-IS (before)




## TO-BE (with the system)

M14 Document Management: TO-BE (with the system)



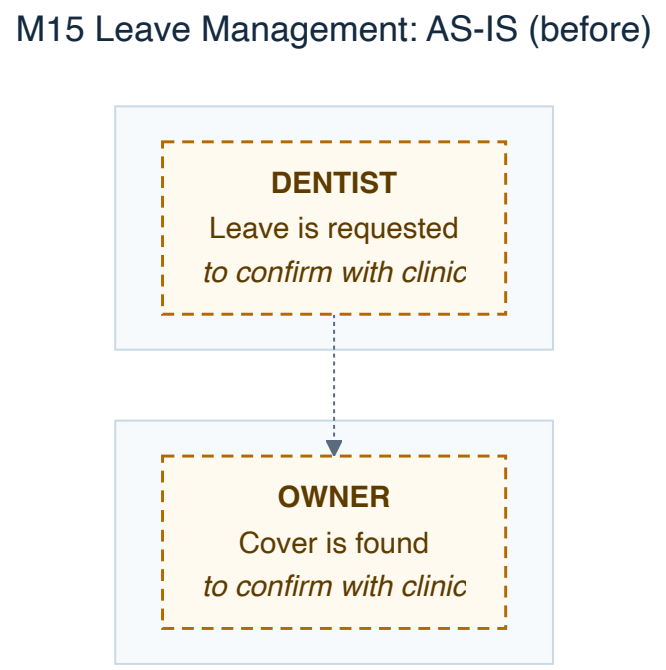

| | Before | Now |
|---|---|---|
| **Steps** | 2 known (1 to confirm) | 3 in the system + 1 planned or mock |
| **Who does it** | Front desk | Patient, System |
| **Steps the system does by itself** | — | 1 |
| **Estimated time** | Not measured (no figure in the interviews) | To be measured in pilot |
| **Errors removed** | — | Lost or shared files |
| **Automation type** | Manual | Rule-based |

# M15 Leave Management

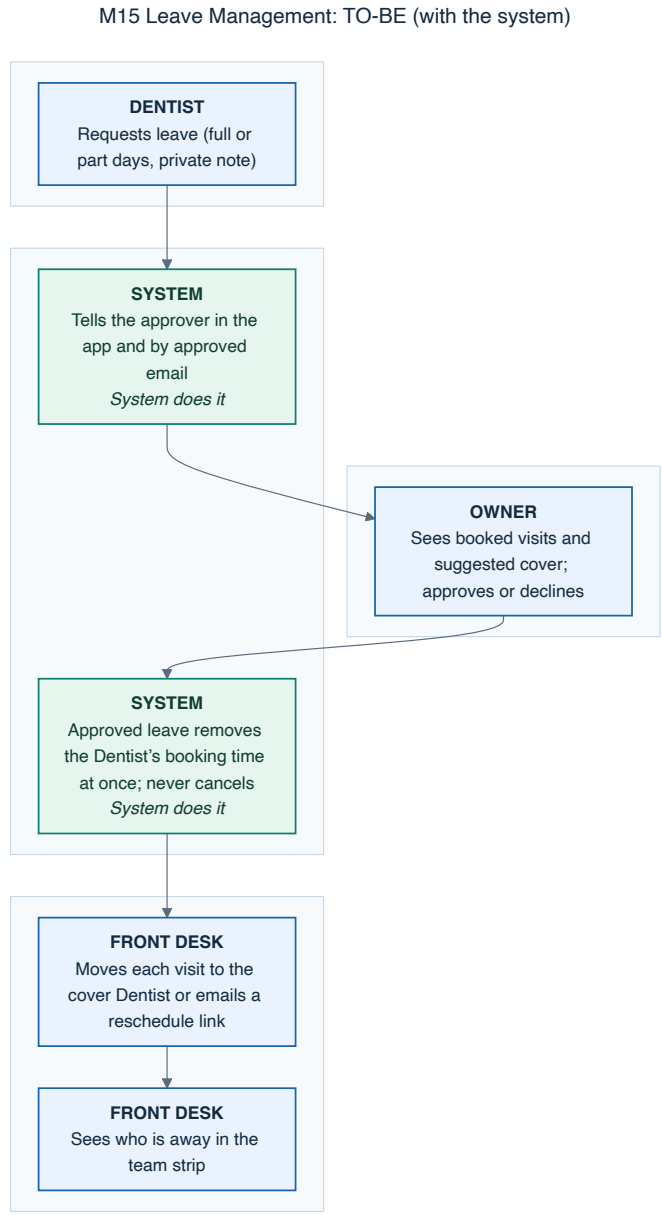
Implemented

AS-IS sources: none in the interview · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

## AS-IS (before)




## TO-BE (with the system)



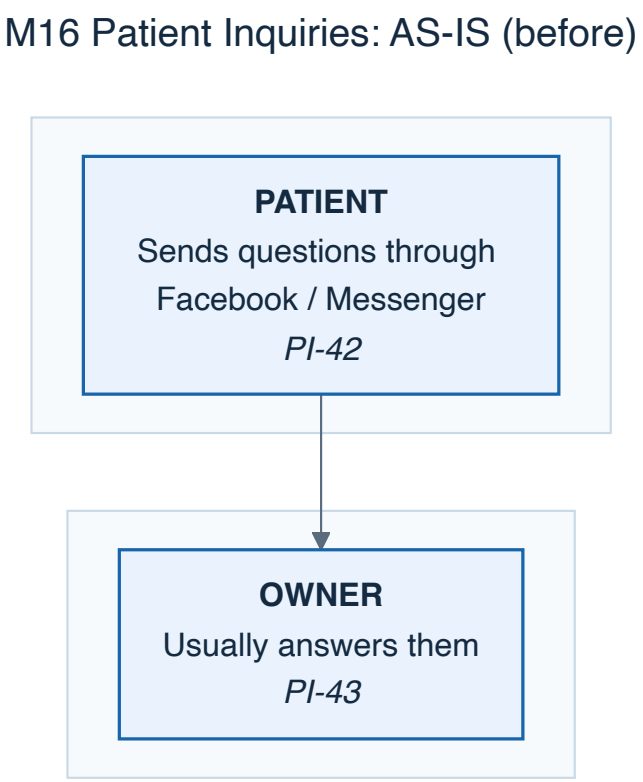

| | Before | Now |
|---|---|---|
| **Steps** | 2 known (2 to confirm) | 6 in the system |
| **Who does it** | Dentist, Owner | Front desk, Dentist, Owner, System |
| **Steps the system does by itself** | — | 2 |
| **Estimated time** | Not measured (no figure in the interviews) | To be measured in pilot |
| **Errors removed** | — | Patients booked with an absent Dentist |
| **Automation type** | Manual | Rule-based |

# M16 Patient Inquiries

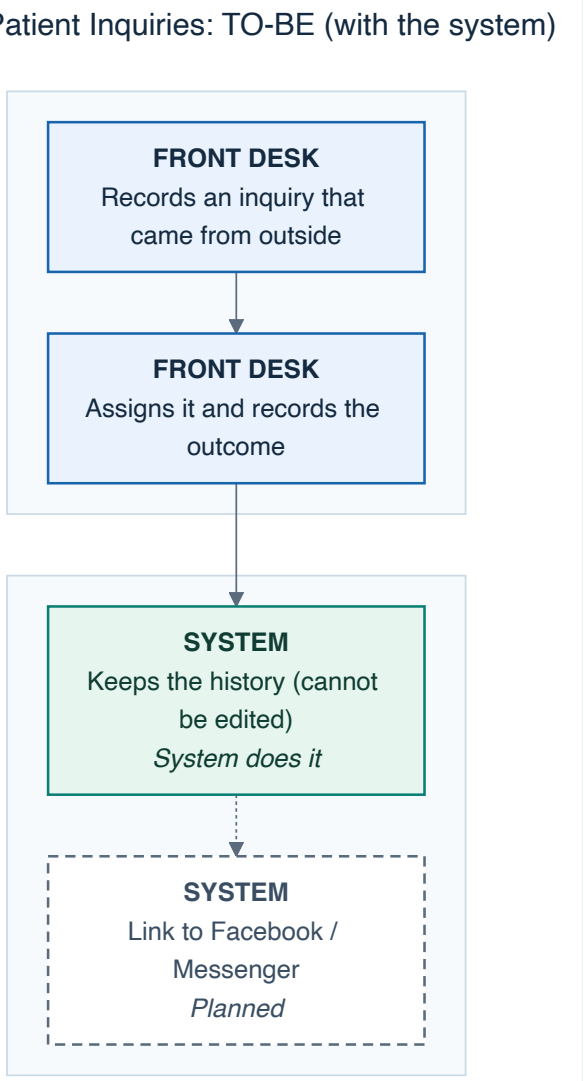
Partial

AS-IS sources: PI-42, PI-43

**AS-IS (before)**




**TO-BE (with the system)**



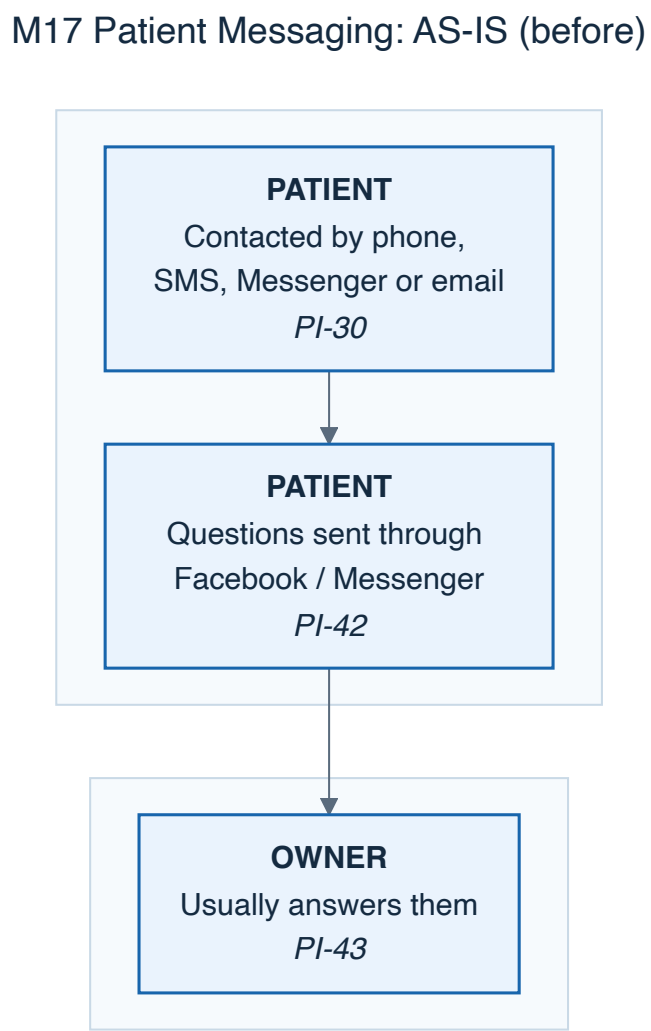

| | Before | Now |
|---|---|---|
| **Steps** | 2 known | 3 in the system + 1 planned or mock |
| **Who does it** | Patient, Owner | Front desk, System |
| **Steps the system does by itself** | — | 1 |
| **Estimated time** | Not measured (no figure in the interviews) | To be measured in pilot |
| **Errors removed** | — | Inquiries forgotten or answered twice |
| **Automation type** | Manual | Rule-based |

# M17 Patient Messaging

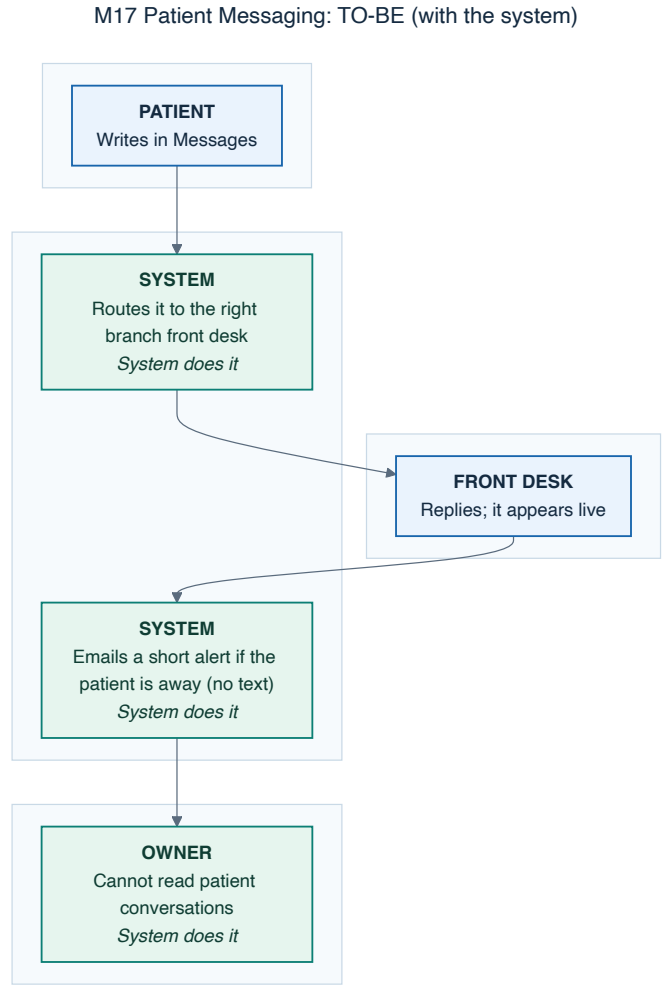
Implemented

AS-IS sources: PI-30, PI-42, PI-43

### AS-IS (before)




### TO-BE (with the system)



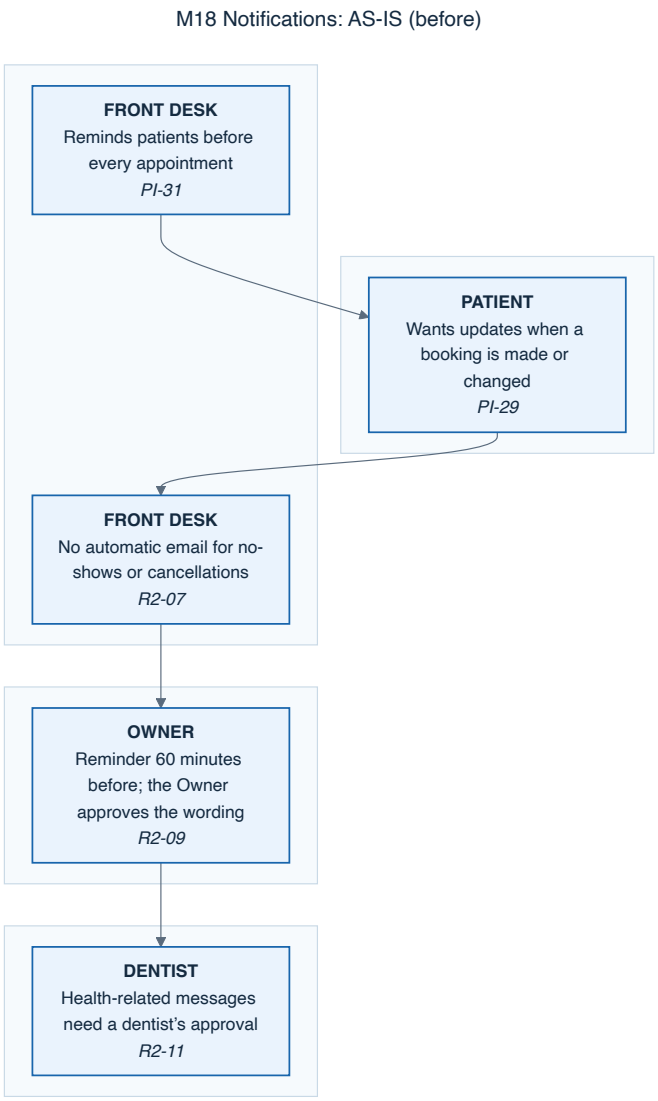

| | Before | Now |
|---|---|---|
| **Steps** | 3 known | 5 in the system |
| **Who does it** | Patient, Owner | Patient, Front desk, Owner, System |
| **Steps the system does by itself** | — | 3 |
| **Estimated time** | Not measured (no figure in the interviews) | To be measured in pilot |
| **Errors removed** | — | Messages to the wrong person or lost in personal accounts |
| **Automation type** | Manual | Event-driven (real-time, alert email) |

# M18 Notifications

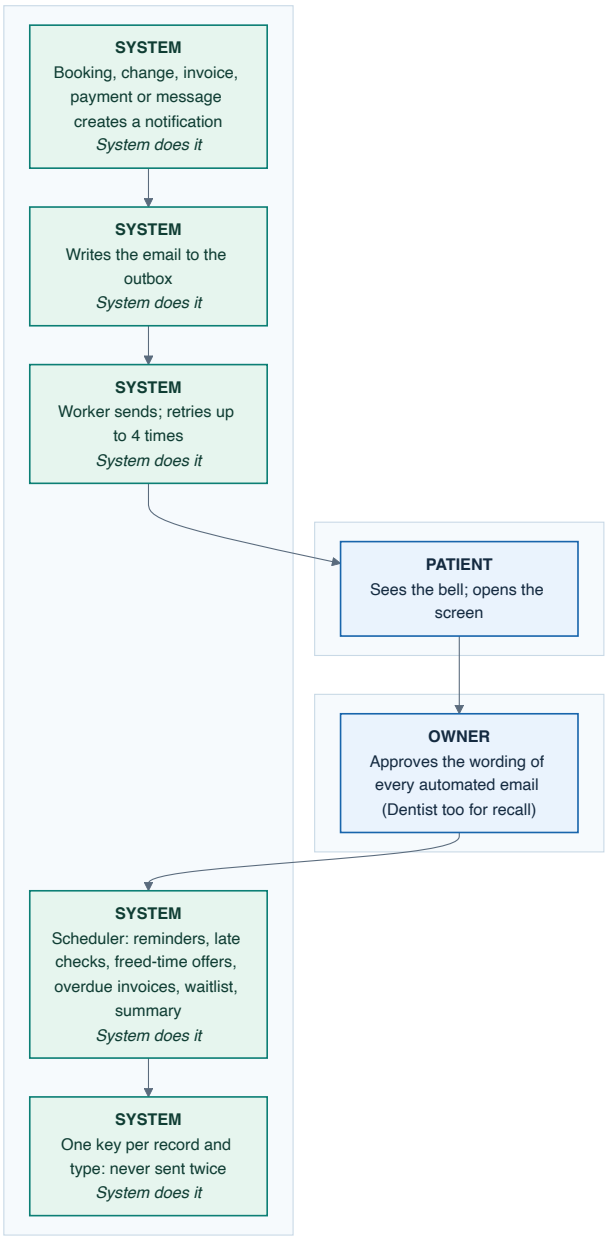
Implemented

AS-IS sources: PI-29, PI-31, R2-07, R2-09, R2-11

## AS-IS (before)




## TO-BE (with the system)



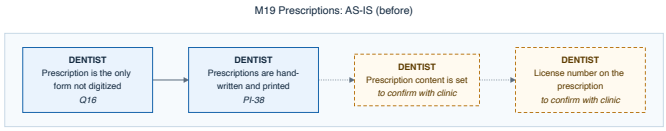

| | Before | Now |
|---|---|---|
| **Steps** | 5 known | 7 in the system |
| **Who does it** | Patient, Front desk, Dentist, Owner | Patient, Owner, System |
| **Steps the system does by itself** | — | 5 |
| **Estimated time** | Not measured (no figure in the interviews) | To be measured in pilot |
| **Errors removed** | — | Missed updates and reminders (PI-31); failed emails are retried; no duplicates |
| **Automation type** | Manual | Event-driven, scheduled (reminders, retries) |

# M19 Prescriptions

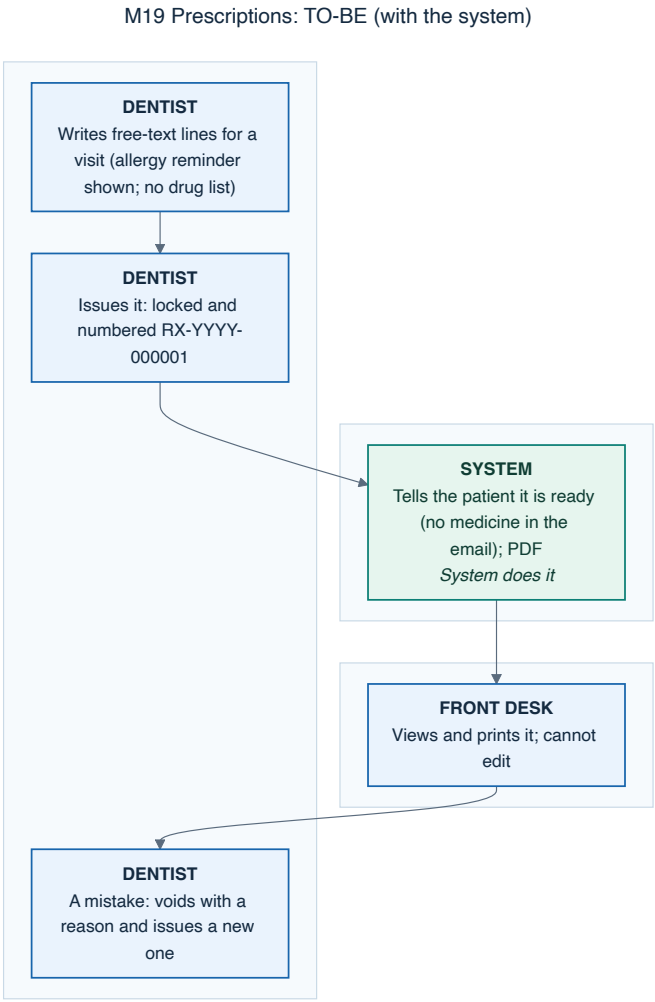
Implemented

AS-IS sources: PI-38, Q16 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

## AS-IS (before)




## TO-BE (with the system)



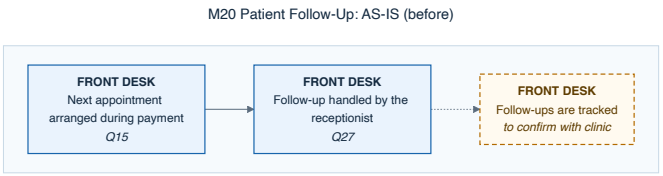

| | Before | Now |
|---|---|---|
| **Steps** | 4 known (2 to confirm) | 5 in the system |
| **Who does it** | Dentist | Front desk, Dentist, System |
| **Steps the system does by itself** | — | 1 |
| **Estimated time** | Not measured (no figure in the interviews) | To be measured in pilot |
| **Errors removed** | — | Lost or illegible paper prescriptions; edits after handing over |
| **Automation type** | Manual | Rule-based (numbering, lock, notice) |

# M20 Patient Follow-Up

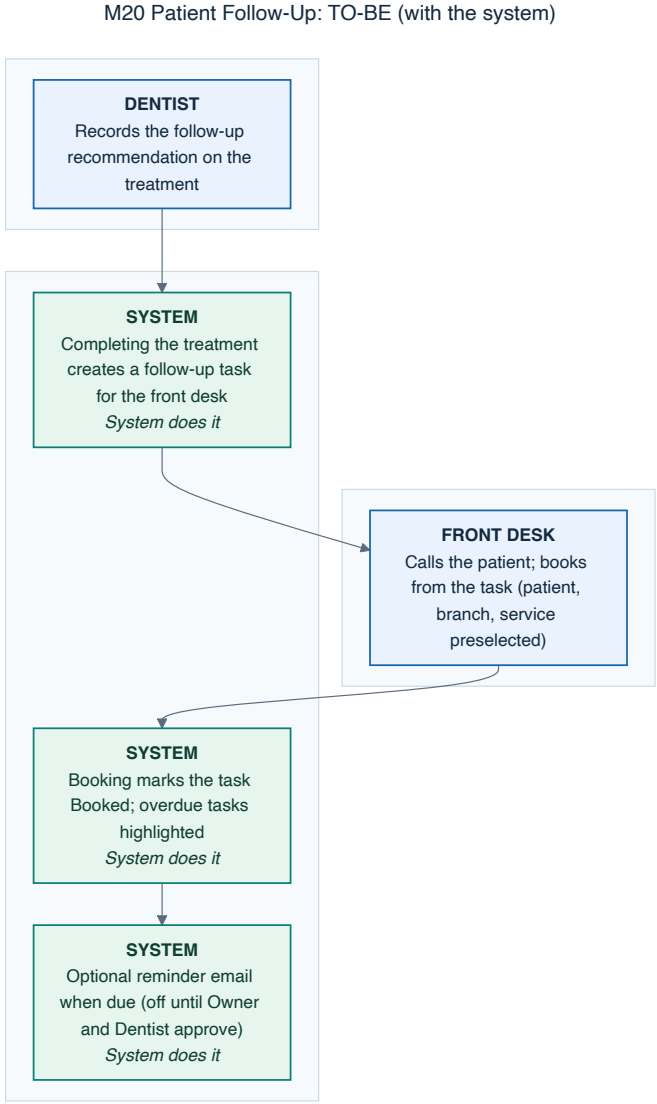
Implemented

AS-IS sources: Q15, Q27 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

## AS-IS (before)




## TO-BE (with the system)



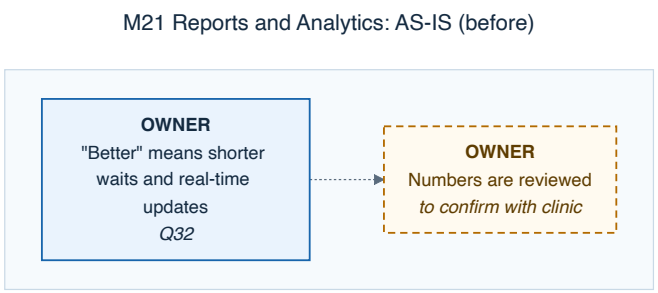

| | Before | Now |
|---|---|---|
| **Steps** | 3 known (1 to confirm) | 5 in the system |
| **Who does it** | Front desk | Front desk, Dentist, System |
| **Steps the system does by itself** | — | 3 |
| **Estimated time** | Not measured (no figure in the interviews) | To be measured in pilot |
| **Errors removed** | — | Follow-ups forgotten when not booked at payment |
| **Automation type** | Manual | Rule-based tasks; optional scheduled reminder (off by default) |

# M21 Reports and Analytics

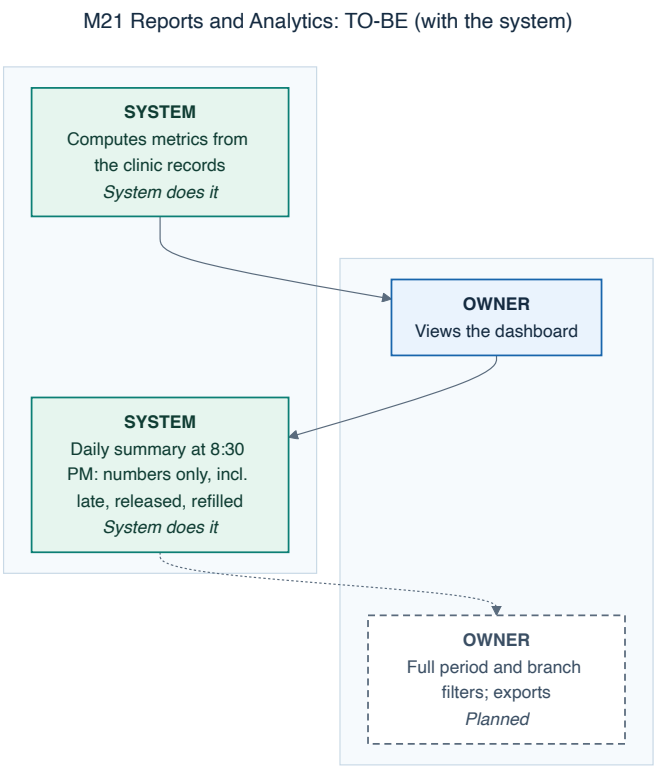
Partial

AS-IS sources: Q32 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

## AS-IS (before)




## TO-BE (with the system)



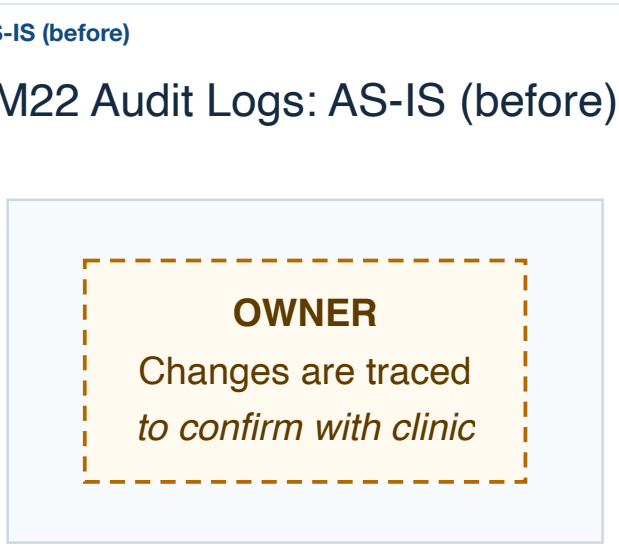

| | Before | Now |
|---|---|---|
| **Steps** | 2 known (1 to confirm) | 3 in the system + 1 planned or mock |
| **Who does it** | Owner | Owner, System |
| **Steps the system does by itself** | — | 2 |
| **Estimated time** | Not measured (no figure in the interviews) | To be measured in pilot |
| **Errors removed** | — | Numbers compiled by hand |
| **Automation type** | Manual | Rule-based |

# M22 Audit Logs

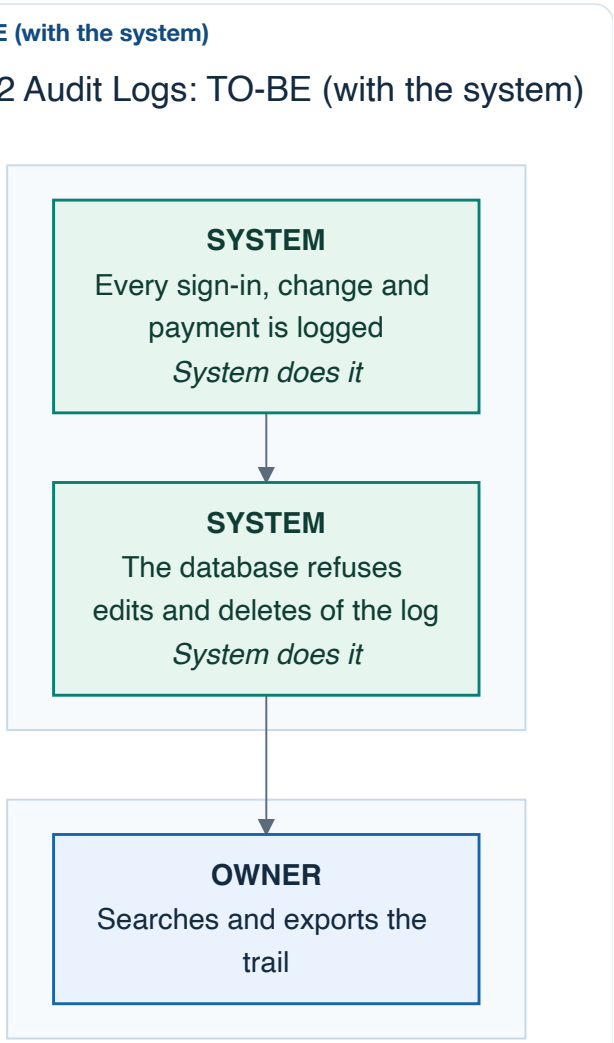
Implemented

AS-IS sources: none in the interview · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

**AS-IS (before)**




**TO-BE (with the system)**



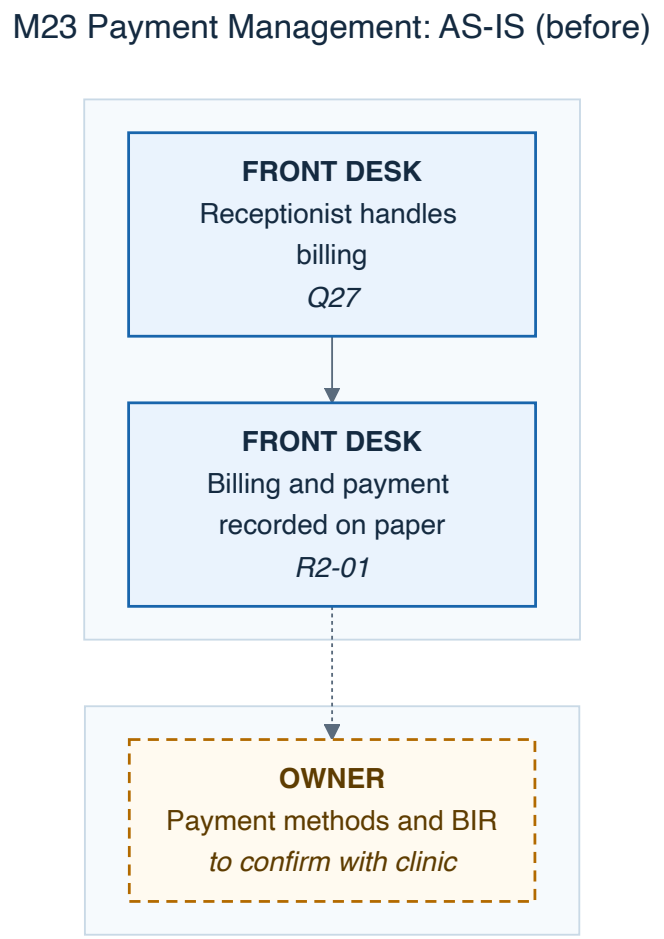

| | Before | Now |
|---|---|---|
| **Steps** | 1 known (1 to confirm) | 3 in the system |
| **Who does it** | Owner | Owner, System |
| **Steps the system does by itself** | — | 2 |
| **Estimated time** | Not measured (no figure in the interviews) | To be measured in pilot |
| **Errors removed** | — | Untraceable changes |
| **Automation type** | Manual | Event-driven |

# M23 Payment Management

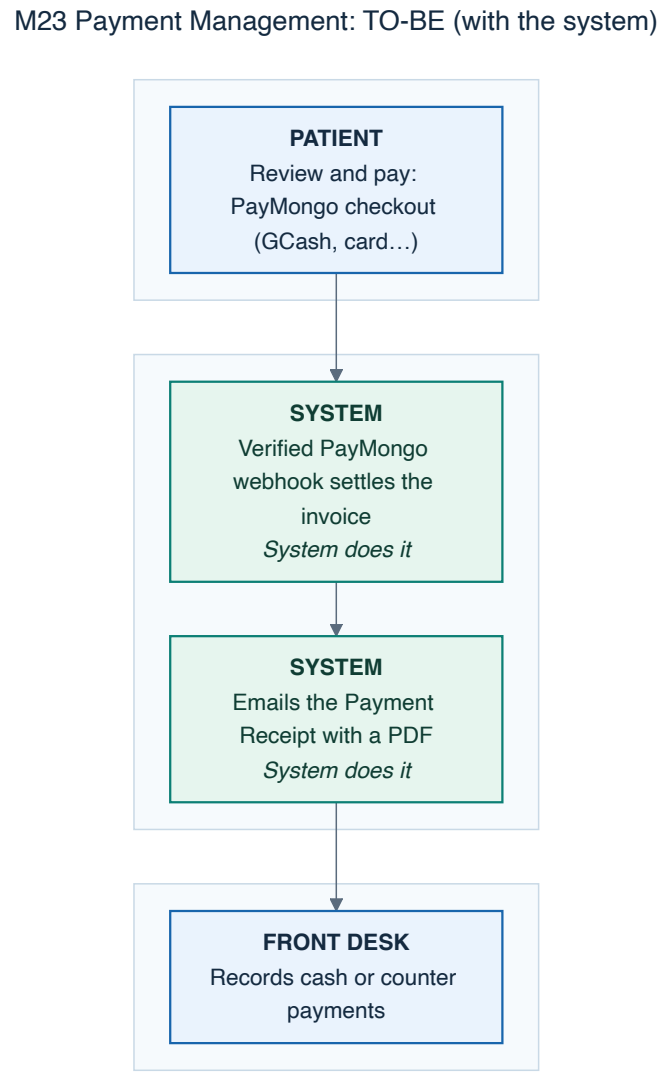
Implemented

AS-IS sources: Q27, R2-01 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

**AS-IS (before)**




**TO-BE (with the system)**



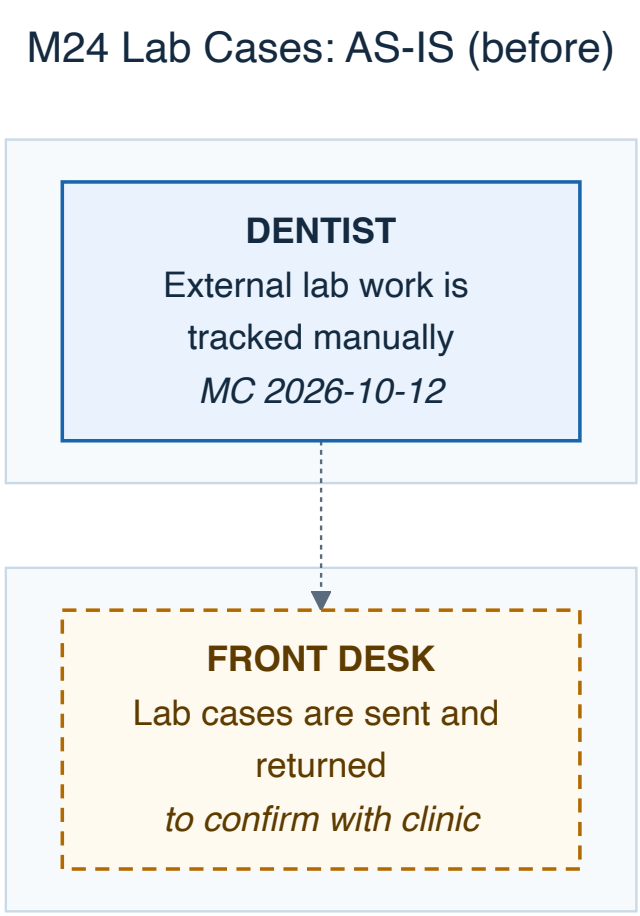

| | Before | Now |
|---|---|---|
| **Steps** | 3 known (1 to confirm) | 4 in the system |
| **Who does it** | Front desk, Owner | Patient, Front desk, System |
| **Steps the system does by itself** | — | 2 |
| **Estimated time** | Not measured (no figure in the interviews) | To be measured in pilot |
| **Errors removed** | — | Payments typed by hand; "paid" without proof |
| **Automation type** | Manual | Integration (PayMongo), event-driven (receipt) |

# M24 Lab Cases

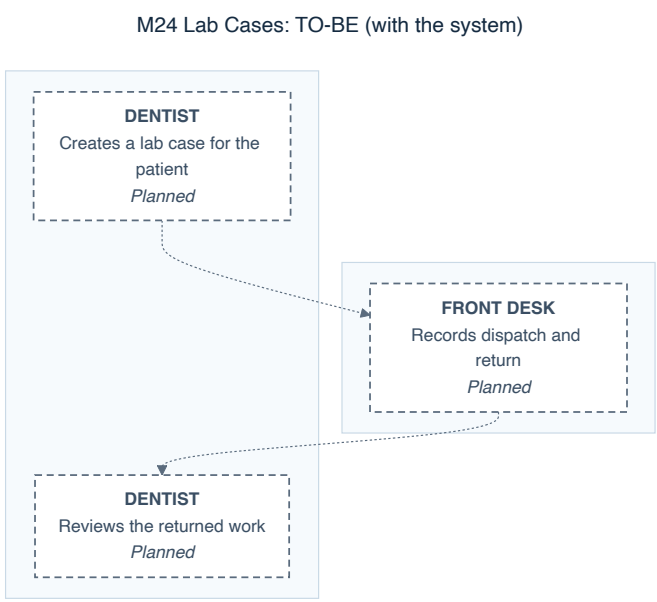
Planned

AS-IS sources: MC 2026-10-12 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

**AS-IS (before)**




**TO-BE (with the system)**



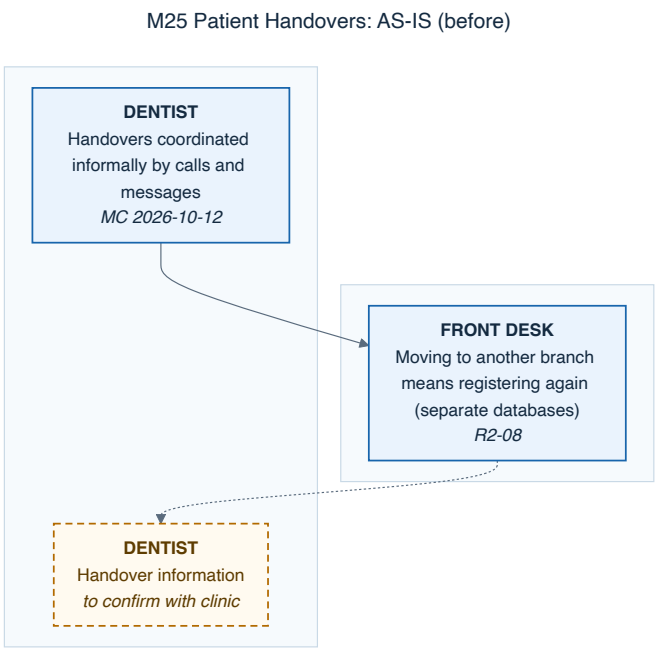

| | Before | Now |
|---|---|---|
| **Steps** | 2 known (1 to confirm) | 0 in the system + 3 planned or mock |
| **Who does it** | Front desk, Dentist | Nobody yet (planned) |
| **Steps the system does by itself** | — | 0 |
| **Estimated time** | Not measured (no figure in the interviews) | To be measured in pilot |
| **Errors removed** | — | Not built yet |
| **Automation type** | Manual | Planned |

# M25 Patient Handovers

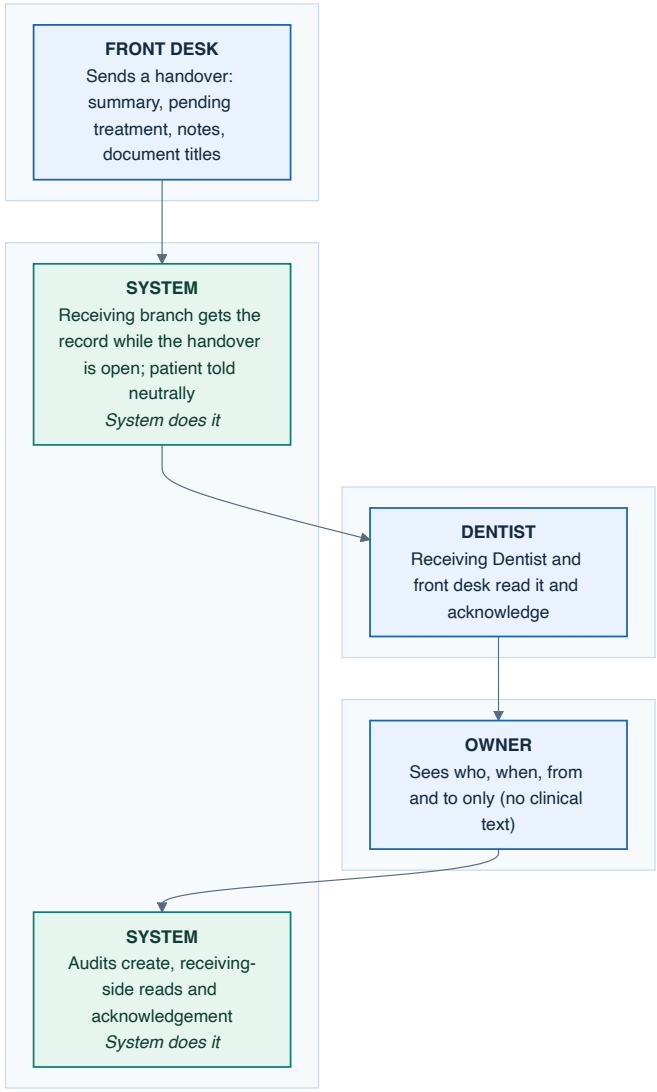
Implemented

AS-IS sources: MC 2026-10-12, R2-08 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

## AS-IS (before)




## TO-BE (with the system)




| | Before | Now |
|---|---|---|
| **Steps** | 3 known (1 to confirm) | 5 in the system |
| **Who does it** | Front desk, Dentist | Front desk, Dentist, Owner, System |
| **Steps the system does by itself** | — | 2 |
| **Estimated time** | Not measured (no figure in the interviews) | To be measured in pilot |
| **Errors removed** | — | Care details lost when a patient moves branch |
| **Automation type** | Manual | Rule-based |

## Limitations and open items

- Planned modules (M3, M10, M12, M15, M24, M25) and mock-only modules (M19, M20) are drawn as designs, not working software.
- Six scheduled automations (reminders, possible no-shows, overdue invoices, recall, waitlist, Owner summary) run with defaults the clinic has not confirmed; recall ships off with no rules. A no-show is only flagged: a person confirms it.
- Online times are limited by the dentists' seeded shifts until the clinic gives real schedules (PI-06).
- Branches 1 and 2 share one phone number (to confirm); branch 3 has no email and an incomplete address.
- Payments run in PayMongo test mode; live payments need the clinic's own PayMongo account.
- Privacy Policy and Terms are drafts; retention periods are not decided (RA 10173).
- Some timestamps are stored 8 hours early (docs/TIMEZONE_PLAN.md); fix planned after the defense.
- 28 questions for the second interview: docs/defense/SECOND_INTERVIEW_QUESTIONS.md. Full list of clinic decisions: docs/defense/OPEN_ITEMS.md.